DEEP RESEARCH DOSSIER (OLD) • DETERMINATIVE V5.3

Sample: `Melinda Ehrle` (`Melinda_Ehrle`) | Reference: GRCh38.p14 | Modality: 33.4x mean depth (Median 30.0x) WGS (panSN GBZ aligned)

---

**AI-Generated Clinical Research Synthesis — Non-Diagnostic Research Use Only**

This report is computationally synthesized using local open-weight models and is intended strictly for exploratory genomic research, variant prioritization, and informational purposes only. It is not an in vitro diagnostic test, does not constitute medical advice or clinical diagnosis, and is not intended for direct clinical management, prognosis, or therapeutic prescription. AlphaGenome Atlas annotations are not intended for clinical or diagnostic purposes; predicted variant impact scores represent computational deep learning estimates.

---

This clinical genomics evidence synthesis evaluates a broad exploratory screening corpus of 1,656 prioritized candidate variants across 1,131 clinical genes derived from a **33.4x mean depth (Median 30.0x)** whole-genome sequencing (WGS) pipeline for Melinda Ehrle. To prevent cognitive bias and diagnostic over-interpretation, this report enforces rigorous domain segregation across **Primary Monogenic Pathogenic Alleles**, **Actionable Pharmacogenomics**, **Secondary Organ-System Modifiers**, and **Endogenous Protective Factors**.

Within this cohort, orthogonal consensus between established human disease databases (ClinVar, OMIM, ClinGen) and biological foundation models (DeepMind AlphaGenome, AlphaMissense, CADD, SpliceAI) identifies **three definitive pathogenic monogenic carrier states** (`POLG p.Gly737Arg`, `ATM c.1236-2del`, and `TAT p.Arg57Ter`), alongside high-impact pharmacogenomic Poor Metabolizer profiles (`CYP2C19 *2/*2`, `CYP3A5 *3/*3`, and `SLCO1B1 *5/*37`).

---

| Gene | Variant | Zygosity | ClinVar | CADD (Phred) | REVEL | AlphaGenome AVI (Phred) | Associated Condition |
|---|---|---|---|---|---|---|---|
| **POLG** | `p.Gly737Arg (c.2209G>C)` | Het (Carrier) | Pathogenic | 26.3 | 0.936 | **30.3 (Cactus)** | Mitochondrial DNA Depletion & Valproate Toxicity Risk (MIM:174763) |
| **ATM** | `c.1236-2del (Splice Acceptor)` | Het (Carrier) | Pathogenic | — | — | — | Hereditary Breast Cancer Susceptibility (MIM:607585) |
| **TAT** | `p.Arg57Ter (c.169C>T)` | Het (Carrier) | Pathogenic | 36.0 | — | **38.8 (Splicing)** | Tyrosinemia Type II Carrier (MIM:276600) |

| Gene | Variant | Zygosity | ClinVar | CADD (Phred) | REVEL | AlphaGenome AVI (Phred) | Associated Condition |
|---|---|---|---|---|---|---|---|
| **VDR** | `c.-1172A>G` | Het (Carrier) | Likely pathogenic | 5.4 | — | 2.1 | Vitamin D Receptor Promoter Regulatory Allele [VCV3336650] |

---

| Gene | Diplotype | Phenotype | High-Risk Medications | Clinical Prescribing Guidance |
|---|---|---|---|---|
| **POLG** | `p.Gly737Arg` | Mitochondrial Carrier | **Sodium Valproate (Depakote)** | **ABSOLUTE CONTRAINDICATION:** Extreme risk of fatal acute hepatic necrosis. Avoid valproate in all forms; select alternative antiepileptic agents. |
| **CYP2C19** | `*2/*2` | Poor Metabolizer | **Clopidogrel, Omeprazole, Citalopram** | Complete loss of CYP2C19 bioactivation. Markedly impaired clopidogrel antiplatelet efficacy; CPIC Level A advises prasugrel or ticagrelor. Adjust PPI dosing. |
| **SLCO1B1** | `*5/*37` | Decreased Function | **Simvastatin, Atorvastatin** | Reduced hepatic OATP1B1 uptake; heightened myopathy risk with simvastatin. CPIC Level A advises lower starting dose or rosuvastatin/pravastatin. |
| **CYP3A5** | `*3/*3` | Poor Metabolizer | **Tacrolimus** | Non-expresser phenotype; requires standard Caucasian starting dosing with therapeutic drug monitoring. |

---

ANALYTICAL CALLSET CONFIDENCE

**98%**

High-depth 33.4x mean depth (Median 30.0x) WGS callset aligned against panSN GBZ pan-genome graph; >99.8% callable SNV precision.

PRIMARY PATHOGENIC CARRIER STATUS (POLG, ATM, TAT)

**95%**

Orthogonal concordance between ClinVar Pathogenic curation records, CADD scores > 26, and AlphaGenome AVI scores > 30.

## CLINICAL DOMAIN GROUPINGS • LANDSCAPE LAYOUT

Organized into distinct clinical categories. Formatted with page breaks and landscape orientation for clean multi-page review.

## 1. Pathogenic & Monogenic Carriers

5 Variants

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | External Accessio |
|---|---|---|---|---|---|---|---|---|---|---|
| **ATM** | c.1236-2del | chr11:108250699 | rs1565381646 | Heterozygous | **Tier1** | Pathogenic/Likely pathogenic | — | — | — | VCV5745 |
| **CDKN2B** | c.*2619C>T | chr9:22003368 | rs1063192 | Heterozygous | Tier3 | Likely pathogenic\|protective | 0.2 | — | 1.3 | VCV8126 |
| **POLG** | p.Gly737Arg | chr15:89323460 | rs121918054 | Heterozygous | **Tier1** | Pathogenic/Likely pathogenic | 26.3 | 0.936 | 30.3 | VCV13513 |
| **TAT** | p.Arg57Ter | chr16:71576247 | rs118203914 | Heterozygous | **Tier1** | Pathogenic | 36.0 | — | 38.8 | VCV402 / |
| **VDR** | c.-1172A>G | chr12:47906043 | rs4516035 | Heterozygous | Tier3 | Likely pathogenic | 5.4 | — | 2.1 | VCV3336 |

## 2. Actionable Pharmacogenomics & Drug Response

66 Variants

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | External Accessio |
|---|---|---|---|---|---|---|---|---|---|---|
| **ABCB1** | c.2065-76T>A | chr7:87544351 | rs1922242 | Heterozygous | Tier3 | drug response | 0.7 | — | 0.8 | VCV8295 |
| **ABCB1** | c.2064+73A>G | chr7:87544750 | rs2235046 | Heterozygous | Tier3 | drug response | 0.5 | — | 5.8 | VCV8287 |
| **ABCB1** | c.1725+38G>A | chr7:87549310 | rs2235013 | Heterozygous | Tier3 | drug response | 6.4 | — | 1.4 | VCV8286 |
| **ABCB1** | c.1554+81C>T | chr7:87549770 | rs2235035 | Heterozygous | Tier3 | drug response | 2.0 | — | 1.1 | VCV8294 |
| **ABCB1** | c.1554+24T>C | chr7:87549827 | rs2235033 | Heterozygous | Tier3 | drug response | 1.8 | — | 0.5 | VCV8294 |
| **ABCB1** | c.1000-44G>A | chr7:87550882 | rs10276036 | Heterozygous | Tier3 | drug response | 0.2 | — | 0.6 | VCV8294 |
| **ABCB1** | c.828-106A>G | chr7:87554038 | rs1922240 | Heterozygous | Tier3 | drug response | 5.1 | — | 2.7 | VCV8286 |
| **ABCB1** | c.827+649A>C | chr7:87560614 | rs868755 | Heterozygous | Tier3 | drug response | 1.0 | — | 0.1 | VCV8294 |
| **ABCB1** | c.703-966G>C | chr7:87562353 | rs13237132 | Heterozygous | Tier3 | drug response | 0.8 | — | 8.7 | VCV8286 |
| **ABCB1** | c.702+280G>A | chr7:87565790 | rs1202170 | Heterozygous | Tier3 | drug response | 0.2 | — | 0.1 | VCV8294 |
| **ABCB1** | c.530+139C>T | chr7:87566646 | rs1202168 | Heterozygous | Tier3 | drug response | 3.1 | — | 6.4 | VCV8286 |
| **ABCB1** | c.338+306C>T | chr7:87569866 | rs1016793 | Heterozygous | Tier3 | drug response | 0.2 | — | 1.5 | VCV8286 |
| **ABCB1** | c.338+224G>T | chr7:87569948 | rs2235021 | Heterozygous | Tier3 | drug response | 0.7 | — | 3.1 | VCV8285 |
| **ABCB1** | c.338+223A>T | chr7:87569949 | rs2235020 | Heterozygous | Tier3 | drug response | 1.2 | — | 1.8 | VCV8285 |
| **ABCB1** | c.286+553T>G | chr7:87584959 | rs7809208 | Heterozygous | Tier3 | drug response | 1.6 | — | 1.2 | VCV8285 |
| **ABCB1** | c.118-118T>G | chr7:87585798 | rs1211152 | Homozygous | Tier3 | drug response | 0.3 | — | 0.7 | VCV8285 |
| **ABCB1** | c.-1A>G | chr7:87600185 | rs2214102 | Homozygous | Tier3 | drug response | 7.0 | — | 5.1 | VCV8293 |
| **ACP1** | c.231+53C>T | chr2:272203 | rs11553746 | Heterozygous | Tier3 | drug response | 14.7 | 0.143 | 24.3 | VCV3900 |
| **ADRB2** | p.Gly16Arg | chr5:148826877 | rs1042713 | Heterozygous | Tier3 | drug response | 17.9 | 0.079 | 19.5 | VCV2259 |
| **APOE** | c.-286T>G | chr19:44905579 | rs405509 | Heterozygous | Tier3 | drug response | 10.2 | — | 5.2 | VCV61995 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | External Accessio |
|---|---|---|---|---|---|---|---|---|---|---|
| **APOE** | c.43+520G>A | chr19:44907187 | rs769450 | Heterozygous | Tier3 | drug response | 4.0 | — | 6.1 | VCV61995 |
| **APOE** | c.*1944T>C | chr19:44911194 | rs439401 | Heterozygous | Tier3 | drug response | 5.6 | — | 7.9 | VCV61995 |
| **ATIC** | c.1503+675T>C | chr2:215347616 | rs4673993 | Heterozygous | Tier3 | drug response | 2.1 | — | 0.8 | VCV2259 |
| **CHI3L1** | p.Arg145Gly | chr1:203183673 | rs880633 | Homozygous | Tier3 | drug response | 23.4 | 0.098 | 18.5 | VCV3900 |
| **COMT** | c.*522G>A | chr22:19969258 | rs165599 | Heterozygous | Tier3 | drug response | 1.8 | — | 4.5 | VCV8289 |
| **COMT** | c.*764C>T | chr22:19969500 | rs165728 | Homozygous | Tier3 | drug response | 1.0 | — | 0.4 | VCV8297 |
| **CYP19A1** | c.145+418G>T | chr15:51242350 | rs727479 | Heterozygous | Tier3 | drug response | 0.0 | — | 1.1 | VCV2581 |
| **CYP2C19** | c.332-23A>G | chr10:94775367 | rs12769205 | Homozygous | Tier3 | drug response | 5.9 | — | 11.3 | VCV6338 |
| **CYP3A5** | c.219-237A>G | chr7:99672916 | rs776746 | Homozygous | Tier3 | association\|drug response\|risk factor | 3.4 | — | 20.6 | VCV5066 |
| **CYP4F2** | p.Val433Met | chr19:15879621 | rs2108622 | Heterozygous | Tier3 | drug response | 24.9 | 0.305 | 13.1 | VCV2259 |
| **DIO1** | c.682-34C>A | chr1:53909897 | rs2235544 | Heterozygous | Tier3 | drug response | 2.3 | — | 3.9 | VCV2573 |
| **HSD3B1** | p.Thr367Asn | chr1:119514623 | rs1047303 | Homozygous | Tier3 | Benign\|drug response | 0.0 | 0.126 | 7.8 | VCV1274 |
| **ITPA** | p.Pro32Thr | chr20:3213196 | rs1127354 | Heterozygous | Tier2 | drug response | 23.0 | 0.312 | 23.5 | VCV1474 |
| **MTHFR** | p.Ala222Val | chr1:11796321 | rs1801133 | Heterozygous | Tier2 | drug response | 27.3 | 0.842 | 25.8 | VCV3520 |
| **NAT2** | p.Gly286Glu | chr8:18400860 | rs1799931 | Heterozygous | Tier2 | drug response | 0.0 | 0.022 | 0.0 | VCV725 / |
| **NPC1L1** | c.-1473T>C | chr7:44542732 | rs2073547 | Heterozygous | Tier3 | drug response | 3.3 | — | 1.8 | VCV18014 |
| **OPRM1** | p.Asn40Asp | chr6:154039662 | rs1799971 | Heterozygous | Tier2 | Uncertain significance\|drug response | 24.1 | 0.104 | 21.5 | VCV9538 |
| **OPRM1** | c.291-2309C>T | chr6:154087517 | rs78937301 | Heterozygous | Tier3 | drug response | 0.0 | — | 0.2 | VCV8284 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | External Accessio |
|---|---|---|---|---|---|---|---|---|---|---|
| **OPRM1** | c.644-83G>C | chr6:154090869 | rs2075572 | Heterozygous | Tier3 | drug response | 0.1 | — | 0.6 | VCV8292 |
| **OPRM1** | c.1165-11281G>A | chr6:154107402 | rs650825 | Homozygous | Tier3 | drug response | 0.2 | — | 0.0 | VCV8283 |
| **OPRM1** | c.1165-11116A>G | chr6:154107567 | rs650245 | Homozygous | Tier3 | drug response | 3.0 | — | 1.2 | VCV8291 |
| **OPRM1** | c.1165-10726del | chr6:154107951 | chr6:154107951 | Homozygous | Tier3 | drug response | — | — | — | VCV8283 |
| **OPRM1** | c.1165-9792A>G | chr6:154108891 | rs623956 | Homozygous | Tier3 | drug response | 0.9 | — | 0.4 | VCV8291 |
| **OPRM1** | c.1165-8425C>G | chr6:154110258 | rs497332 | Homozygous | Tier3 | drug response | 3.4 | — | 4.9 | VCV8291 |
| **OPRM1** | c.1165-8253G>A | chr6:154110430 | rs606545 | Homozygous | Tier3 | drug response | 5.5 | — | 4.4 | VCV8283 |
| **SCN1A** | c.603-91G>A | chr2:166053034 | rs3812718 | Heterozygous | Tier2 | drug response | 20.4 | — | 23.1 | VCV12897 |
| **SLC19A1** | p.His27Arg | chr21:45537880 | rs1051266 | Homozygous | Tier2 | drug response | 7.5 | 0.192 | 9.0 | VCV15758 |
| **SLC29A3** | p.Ser158Phe | chr10:71351651 | rs780668 | Homozygous | Tier2 | Benign\|drug response | 23.5 | 0.286 | 23.6 | VCV13034 |
| **SLCO1B1** | p.Val174Ala | chr12:21178615 | rs4149056 | Heterozygous | Tier2 | drug response | 25.1 | 0.221 | 24.8 | VCV3734 |
| **TNF** | c.-488G>A | chr6:31575254 | rs1800629 | Heterozygous | Tier3 | drug response | 3.1 | — | 4.8 | VCV2259 |
| **UGT2B7** | c.-161T>C | chr4:69096360 | rs7668258 | Heterozygous | Tier3 | drug response | 0.3 | — | 0.5 | VCV8282 |
| **UGT2B7** | c.-138G>A | chr4:69096383 | rs73823859 | Heterozygous | Tier3 | drug response | 0.1 | — | 14.9 | VCV8282 |
| **UGT2B7** | c.721+201A>C | chr4:69097442 | rs12647681 | Heterozygous | Tier3 | drug response | 0.6 | — | 8.2 | VCV8290 |
| **UGT2B7** | c.721+209A>G | chr4:69097450 | rs12647682 | Heterozygous | Tier3 | drug response | 0.2 | — | 10.0 | VCV8290 |
| **UGT2B7** | c.721+272T>C | chr4:69097513 | rs12648462 | Heterozygous | Tier3 | drug response | 4.7 | — | 5.3 | VCV8290 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | External Accessio |
|---|---|---|---|---|---|---|---|---|---|---|
| **UGT2B7** | `c.722-181C>T` | `chr4:69098359` | `rs73823860` | Heterozygous | Tier3 | drug response | 0.1 | — | 0.1 | VCV8290 |
| **UGT2B7** | `c.722-78T>C` | `chr4:69098462` | `rs7439326` | Heterozygous | Tier3 | drug response | 1.8 | — | 4.5 | VCV8290 |
| **UGT2B7** | `c.722-49A>G` | `chr4:69098491` | `rs7438244` | Heterozygous | Tier3 | drug response | 0.5 | — | 6.7 | VCV8282 |
| **UGT2B7** | `p.Tyr268His` | `chr4:69098620` | `rs7439366` | Heterozygous | Tier3 | drug response | 6.5 | 0.066 | 10.9 | VCV8282 |
| **UGT2B7** | `p.Ala285=` | `chr4:69098673` | `rs34661811` | Heterozygous | Tier3 | drug response | 4.4 | — | 18.0 | VCV8282 |
| **UGT2B7** | `p.Leu353=` | `chr4:69107231` | `rs4292394` | Heterozygous | Tier3 | drug response | 0.1 | — | 2.3 | VCV8282 |
| **UGT2B7** | `c.1090+64A>T` | `chr4:69107326` | `rs4337789` | Heterozygous | Tier3 | drug response | 0.1 | — | 2.6 | VCV8282 |
| **UGT2B7** | `p.Ala397=` | `chr4:69108203` | `rs57913007` | Heterozygous | Tier3 | drug response | 0.3 | — | 0.9 | VCV8282 |
| **UGT2B7** | `c.*296T>C` | `chr4:69113032` | `rs6851533` | Heterozygous | Tier3 | drug response | 6.7 | — | 4.9 | VCV8281 |
| **VKORC1** | `c.283+837T>C` | `chr16:31092475` | `rs2359612` | Homozygous | Tier2 | drug response | 0.5 | — | 12.6 | VCV2259 |
| **XRCC1** | `p.Gln399Arg` | `chr19:43551574` | `rs25487` | Heterozygous | Tier3 | drug response | 14.4 | 0.072 | 8.8 | VCV2259 |

## 3. Cardiovascular & Channelopathy Modifiers

201 Variants

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **AARS2** | c.*448G>T | chr6:44300099 | rs565803591 | Heterozygous | Tier2 | Uncertain significance | 1.0 | — | 3.9 |
| **ABCG5** | p.Ala429Thr | chr2:43823952 | rs779125742 | Heterozygous | Tier2 | Uncertain significance | 21.3 | 0.383 | 14.3 |
| **ABCG8** | p.Thr401Ser | chr2:43872296 | rs144200355 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 13.9 | 0.083 | 10.4 |
| **ADAMTS10** | p.Arg62Gln | chr19:8605262 | rs62621197 | Heterozygous | Tier3 | Benign | 23.2 | 0.018 | 16.2 |
| **APOB** | p.Ser4338Asn | chr2:21002409 | rs1042034 | Homozygous | Tier2 | Benign | 0.8 | 0.055 | 4.3 |
| **APOB** | p.Thr2515= | chr2:21009323 | rs693 | Heterozygous | Tier2 | Benign | 5.4 | — | 10.5 |
| **APOB** | p.Ile2313Val | chr2:21009931 | rs584542 | Homozygous | Tier2 | Benign | 0.1 | 0.014 | 3.6 |
| **APOB** | p.Asp2312= | chr2:21009932 | rs1041968 | Heterozygous | Tier2 | Benign | 0.5 | — | 7.0 |
| **APOB** | c.3843-77T>C | chr2:21013610 | rs488329 | Homozygous | Tier2 | Benign | 6.3 | — | 3.1 |
| **APOB** | p.Leu1060= | chr2:21016593 | rs72653073 | Heterozygous | **Tier1** | Conflicting classifications of pathogenicity | 9.1 | — | 11.5 |
| **APOB** | p.Thr98Ile | chr2:21041028 | rs1367117 | Homozygous | Tier2 | Benign/Likely benign | 22.8 | 0.093 | 18.4 |
| **APOB** | p.Leu12_Leu14del | chr2:21043903 | rs17240441 | Homozygous | Tier2 | Benign/Likely benign | — | — | — |
| **ARID1A** | p.Arg1906Gln | chr1:26779615 | rs41303631 | Heterozygous | Tier2 | Benign/Likely benign | 20.9 | 0.125 | 20.9 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **ATN1** | p.Lys236Gln | chr12:6935973 | rs200232074 | Heterozygous | Tier2 | Benign/Likely benign | 17.8 | 0.051 | 20.1 |
| **ATP5F1A** | c.-32G>T | chr18:46098263 | rs367840441 | Heterozygous | Tier3 | Benign | 17.3 | — | 20.6 |
| **CACNA1A** | p.Gln2325dup | chr19:13207859 | chr19:13207859 | Heterozygous | Tier3 | Benign/Likely benign | — | — | — |
| **CACNA1D** | c.140-65995C>G | chr3:53431157 | rs1156693798 | Heterozygous | Tier2 | Not reviewed | 10.5 | — | 7.8 |
| **CACNA1S** | c.4113+7T>C | chr1:201050977 | rs6702590 | Heterozygous | Tier2 | Benign | 3.1 | — | 0.1 |
| **CACNA1S** | p.Leu522= | chr1:201077934 | rs4915476 | Homozygous | Tier2 | Benign | 9.8 | — | 16.1 |
| **CACNA1S** | p.Gly517= | chr1:201077947 | rs4915477 | Homozygous | Tier2 | Benign | 3.4 | — | 0.0 |
| **CACNA1S** | p.Ala69Gly | chr1:201110216 | rs12406479 | Heterozygous | Tier2 | Benign | 24.4 | 0.371 | 25.0 |
| **CACNA1S** | c.-231A>G | chr1:201112570 | rs4506487 | Homozygous | Tier2 | Benign | 8.8 | — | 9.7 |
| **CASR** | p.Asp398Asn | chr3:122262227 | rs201177696 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 21.8 | 0.219 | 14.4 |
| **CASZ1** | c.-234+4244dup | chr1:10792322 | chr1:10792322 | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **CDKN1C** | c.-11+61G>A | chr11:2885573 | rs188494894 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 15.4 | — | 16.9 |
| **CHST14** | p.Gln349Pro | chr15:40472259 | rs1463514561 | Heterozygous | **Tier1** | Uncertain significance | 24.3 | 0.540 | 22.9 |
| **CHST3** | c.*346C>T | chr10:72008817 | rs76094341 | Heterozygous | Tier2 | Uncertain significance | 13.5 | — | 10.5 |
| **COL1A1** | p.Ala1075Thr | chr17:50188134 | rs1800215 | Heterozygous | Tier3 | Benign/Likely benign | 20.9 | — | 20.2 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **CPLANE1** | p.Asp1135Val | chr5:37201694 | rs780763762 | Heterozygous | Tier2 | Not reviewed | 19.0 | 0.173 | 6.5 |
| **CPLANE1** | p.Asp1135Tyr | chr5:37201695 | rs747635859 | Heterozygous | Tier2 | Not reviewed | 20.8 | 0.140 | 7.8 |
| **CREBBP** | p.Leu551Ile | chr16:3781229 | rs61753381 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 22.5 | 0.382 | 22.4 |
| **CRPPA** | p.Ser275= | chr7:16301431 | rs766874330 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 9.9 | — | 15.8 |
| **DCC** | c.91+22042A>G | chr18:52362920 | rs8095741 | Heterozygous | Tier2 | Not reviewed | 4.1 | — | 2.7 |
| **DHX37** | p.Glu267Lys | chr12:124977430 | rs33931896 | Heterozygous | Tier3 | Benign | 24.6 | 0.367 | 22.3 |
| **DMD** | c.94-60270G>C | chrX:32910090 | rs1923093 | Homozygous | Tier2 | Not reviewed | 1.1 | — | 5.4 |
| **DMD** | c.32-76315T>A | chrX:33096515 | rs4829276 | Heterozygous | Tier2 | Not reviewed | 1.0 | — | 0.6 |
| **DNAAF1** | p.Asp435Asn | chr16:84170131 | rs149158199 | Heterozygous | Tier3 | Benign/Likely benign | 8.1 | 0.071 | 7.3 |
| **DNAAF11** | c.1140+8154A>G | chr8:132602012 | rs4520144 | Heterozygous | Tier2 | Not reviewed | 1.8 | — | 1.8 |
| **DNAAF2** | p.Gly411Arg | chr14:49633919 | rs1883243941 | Heterozygous | Tier2 | Uncertain significance | 7.3 | 0.006 | 6.1 |
| **DNAH1** | c.3694-352G>T | chr3:52356262 | rs115019178 | Heterozygous | Tier2 | Not reviewed | 11.8 | — | 19.8 |
| **DNAH1** | p.Arg3272Cys | chr3:52391251 | rs200784189 | Heterozygous | Tier2 | Uncertain significance | 28.4 | 0.297 | 25.2 |
| **DNAH11** | p.Thr400Ile | chr7:21570073 | rs72655982 | Heterozygous | Tier2 | Benign/Likely benign | 2.0 | 0.037 | 9.8 |
| **DNAH11** | p.Ala568Thr | chr7:21582013 | rs72655988 | Heterozygous | Tier3 | Benign/Likely benign | 21.8 | 0.132 | 19.9 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **DNAH11** | c.8674-227A>G | chr7:21749451 | rs111889929 | Heterozygous | Tier2 | Likely benign | 2.5 | — | 2.9 |
| **DNAH11** | p.Ala4315Thr | chr7:21894893 | rs72658825 | Heterozygous | Tier3 | Benign | 2.2 | 0.018 | 13.0 |
| **DYNC2H1** | c.366+39T>G | chr11:103113746 | rs772875964 | Heterozygous | Tier2 | Not reviewed | 14.7 | — | 21.9 |
| **DYNC2I1** | p.Pro923Gln | chr7:158934539 | rs73167274 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 25.0 | 0.123 | 24.0 |
| **EBP** | c.301+446A>T | chrX:48524518 | rs62602501 | Heterozygous | Tier2 | Not reviewed | 0.7 | — | 0.0 |
| **EPAS1** | c.*111A>G | chr2:46384771 | rs62137032 | Heterozygous | Tier3 | Benign | 15.3 | — | 13.6 |
| **EVC2** | p.Arg687His | chr4:5622978 | rs144420242 | Heterozygous | Tier2 | Uncertain significance | 16.6 | 0.262 | 11.0 |
| **FANCA** | c.1007-765G>T | chr16:89793312 | rs144352354 | Heterozygous | Tier2 | Not reviewed | 0.6 | — | 8.1 |
| **FANCM** | p.Asn1253Ser | chr14:45176512 | rs45604036 | Heterozygous | Tier3 | Benign/Likely benign | 0.1 | 0.029 | 1.4 |
| **FANCM** | p.Asn1876Ser | chr14:45196458 | rs45557033 | Heterozygous | Tier3 | Benign/Likely benign | 1.0 | 0.019 | 6.9 |
| **FASTKD2** | c.*2960T>C | chr2:206794762 | rs147381048 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 2.5 | — | 2.5 |
| **FBN2** | p.Ile243Thr | chr5:128464822 | rs117524265 | Heterozygous | Tier2 | Benign/Likely benign | 23.3 | 0.573 | 20.5 |
| **FIG4** | c.2546+2077T>G | chr6:109798928 | rs4330563 | Homozygous | Tier2 | Not reviewed | 5.1 | — | 3.9 |
| **FIG4** | c.2547-9862A>C | chr6:109815226 | rs4388322 | Homozygous | Tier2 | Not reviewed | 0.7 | — | 2.2 |
| **FIG4** | c.2547-38009T>C | chr6:109839813 | rs9400325 | Homozygous | Tier2 | Not reviewed | 0.7 | — | 0.5 |
| **FIG4** | c.2547-18770T>C | chr6:109859052 | rs9400330 | Homozygous | Tier2 | Not reviewed | 2.3 | — | 4.4 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **FIG4** | c.2547-16414T>G | chr6:109861408 | rs9481016 | Homozygous | Tier2 | Not reviewed | 3.2 | — | 5.2 |
| **FRAS1** | c.2423-391C>T | chr4:78363122 | rs2100484 | Heterozygous | Tier2 | Not reviewed | 2.1 | — | 1.8 |
| **FRAS1** | c.2722+2111G>A | chr4:78366165 | rs11936250 | Heterozygous | Tier2 | Not reviewed | 0.4 | — | 2.6 |
| **FRAS1** | p.Arg3269Gln | chr4:78511299 | rs61729366 | Heterozygous | Tier2 | Benign/Likely benign | 32.0 | — | 24.1 |
| **FRAS1** | c.11298+469G>A | chr4:78537669 | rs7664814 | Homozygous | Tier2 | Not reviewed | 0.4 | — | 0.8 |
| **FREM2** | c.*224G>A | chr13:38881011 | rs575095989 | Heterozygous | Tier2 | Uncertain significance | 1.5 | — | 3.6 |
| **FREM2** | c.*1321G>A | chr13:38882108 | rs78527665 | Heterozygous | Tier2 | Uncertain significance | 2.5 | — | 11.5 |
| **GBA1** | p.Thr408Met | chr1:155236246 | rs75548401 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 21.3 | 0.731 | 20.0 |
| **GJB4** | p.Arg124Gln | chr1:34761625 | rs140996335 | Heterozygous | Tier2 | Likely benign | 22.1 | 0.494 | 16.8 |
| **GLI2** | p.Arg1526His | chr2:120990542 | rs138987487 | Heterozygous | Tier2 | Benign/Likely benign | 29.8 | 0.256 | 22.4 |
| **GLI3** | c.367+28433C>G | chr7:42119793 | rs1265401266 | Heterozygous | Tier2 | Not reviewed | 19.4 | — | 18.4 |
| **GPC6** | c.877+107375del | chr13:94135255 | chr13:94135255 | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **GTF2IRD2** | c.749-149C>T | chr7:74811101 | rs71270899 | Homozygous | Tier2 | Not reviewed | 3.5 | — | 3.7 |
| **GUSB** | p.Trp627Leu | chr7:65960973 | rs1236992554 | Heterozygous | **Tier1** | Not reviewed | 22.2 | 0.560 | 24.9 |
| **GYG1** | c.7+313G>T | chr3:148991960 | rs1000405725 | Heterozygous | Tier2 | Not reviewed | 8.8 | — | 22.1 |
| **HLA-B** | p.Arg180Gly | chr6:31356248 | rs9266144 | Homozygous | Tier2 | Not reviewed | 0.0 | 0.251 | 10.6 |
| **HLA-B** | p.Leu105Pro | chr6:31356717 | rs9266161 | Heterozygous | **Tier1** | Not reviewed | 22.5 | 0.205 | 20.0 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **HLA-B** | p.Leu105Val | chr6:31356718 | rs9266162 | Heterozygous | Tier2 | Not reviewed | 8.9 | 0.188 | 9.0 |
| **HLA-B** | p.Ala95Thr | chr6:31356748 | rs1131213 | Homozygous | Tier2 | Not reviewed | 13.9 | 0.179 | 6.4 |
| **HLA-B** | p.Gln94His | chr6:31356749 | rs1131212 | Homozygous | Tier2 | Not reviewed | 7.1 | 0.244 | 11.3 |
| **HLA-DRB1** | p.Thr262Arg | chr6:32580249 | rs9269744 | Heterozygous | Tier2 | Not reviewed | 25.3 | 0.224 | 19.7 |
| **HLA-DRB1** | c.371-270T>G | chr6:32582108 | rs28723975 | Heterozygous | Tier2 | Not reviewed | 6.4 | — | 11.3 |
| **HLA-DRB1** | c.371-297A>G | chr6:32582135 | rs28723976 | Heterozygous | **Tier1** | Not reviewed | 7.4 | — | 12.9 |
| **HLA-DRB1** | c.371-396T>C | chr6:32582234 | rs28723982 | Heterozygous | Tier2 | Not reviewed | 0.4 | — | 2.1 |
| **HLA-DRB1** | c.371-415G>A | chr6:32582253 | rs28723984 | Heterozygous | Tier2 | Not reviewed | 5.1 | — | 10.3 |
| **HLA-DRB1** | p.Tyr107Ter | chr6:32584158 | rs11554463 | Heterozygous | Tier2 | Not reviewed | 28.0 | — | 27.2 |
| **HLA-DRB1** | p.Tyr107Phe | chr6:32584159 | rs17879620 | Heterozygous | Tier2 | Not reviewed | 0.1 | 0.146 | 0.0 |
| **HLA-DRB1** | p.Tyr107Asp | chr6:32584160 | rs17884043 | Heterozygous | Tier2 | Not reviewed | 16.5 | 0.132 | 8.6 |
| **HLA-DRB1** | p.Ala103Pro | chr6:32584172 | rs16822805 | Heterozygous | **Tier1** | Uncertain significance | 0.0 | 0.209 | 20.2 |
| **HLA-DRB1** | p.Ala102Gly | chr6:32584174 | rs17878857 | Heterozygous | Tier2 | Not reviewed | 12.2 | 0.311 | 11.4 |
| **HLA-DRB1** | p.Ala102GlyfsTer26 | chr6:32584175 | chr6:32584175 | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **HLA-DRB1** | p.Gln99His | chr6:32584182 | rs17879599 | Heterozygous | Tier2 | Not reviewed | 0.6 | 0.217 | 11.3 |
| **HLA-DRB1** | p.Gln99Glu | chr6:32584184 | rs17881965 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.281 | 2.3 |
| **HLA-DRB1** | p.Pro40Ser | chr6:32584361 | rs9269955 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.034 | 0.0 |
| **HLA-DRB1** | p.Pro40Ala | chr6:32584361 | rs9269955 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.028 | 0.2 |
| **HLA-DRB1** | p.Gln39Ter | chr6:32584364 | rs9269957 | Heterozygous | Tier2 | Uncertain significance | 26.8 | — | 28.2 |
| **HLA-DRB1** | c.101-43C>T | chr6:32584421 | rs28724109 | Heterozygous | Tier2 | Not reviewed | 6.7 | — | 19.2 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **HLA-DRB1** | p.Ala14Val | chr6:32589702 | rs9270302 | Heterozygous | Tier2 | Not reviewed | 15.4 | 0.177 | 12.1 |
| **HSPG2** | c.9710-10G>A | chr1:21839560 | rs369434104 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 0.4 | — | 0.6 |
| **HSPG2** | p.Arg257Trp | chr1:21887609 | rs535956121 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 0.3 | 0.203 | 0.8 |
| **HYDIN** | p.Arg4513His | chr16:70834028 | rs201554059 | Heterozygous | Tier2 | Likely benign | 22.4 | 0.213 | 18.0 |
| **HYDIN** | p.Thr690Ala | chr16:71067297 | rs10744982 | Heterozygous | Tier2 | Not reviewed | 24.9 | 0.082 | 21.7 |
| **IKBKG** | c.1056-6C>T | chrX:154563953 | rs5945206 | Homozygous | Tier2 | Uncertain significance | 10.5 | — | 7.3 |
| **IQSEC2** | p.Ala1470Thr | chrX:53234278 | rs191886831 | Heterozygous | Tier3 | Benign | 14.2 | 0.028 | 14.8 |
| **KDM6A** | c.384+5299del | chrX:44980000 | chrX:44980000 | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **KIF1B** | p.Ser1427Asn | chr1:10361801 | rs755113137 | Heterozygous | Tier2 | Uncertain significance | 25.9 | 0.236 | 24.2 |
| **KIF20A** | p.Pro839Leu | chr5:138187256 | rs3172747 | Heterozygous | Tier3 | Benign | 23.9 | 0.163 | 22.9 |
| **KLHL24** | c.-62+617T>C | chr3:183644159 | rs9877526 | Homozygous | Tier2 | Not reviewed | 6.0 | — | 0.7 |
| **KMT2C** | c.4661-778C>A | chr7:152188625 | rs13242182 | Heterozygous | Tier2 | Not reviewed | 1.1 | — | 0.2 |
| **LDLRAP1** | p.Ile132= | chr1:25557204 | rs369497122 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 2.2 | — | 0.4 |
| **LMBRD1** | p.Leu135fs | chr6:69749975 | chr6:69749975 | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **MAPRE2** | p.Arg8fs | chr18:34979998 | chr18:34979998 | Heterozygous | Tier2 | Not reviewed | — | — | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **MC2R** | c.*992A>T | chr18:13883633 | rs897474751 | Heterozygous | Tier2 | Uncertain significance | 4.4 | — | 0.1 |
| **MRPS22** | c.-71-65596T>G | chr3:139278360 | rs78362422 | Heterozygous | Tier2 | Not reviewed | 2.5 | — | 1.3 |
| **MYH11** | c.-18+2381G>C | chr16:15854560 | rs9933232 | Heterozygous | Tier2 | Not reviewed | 0.5 | — | 1.6 |
| **MYL2** | c.-52-2913T>C | chr12:110923494 | rs12828640 | Heterozygous | Tier2 | Not reviewed | 5.4 | — | 4.6 |
| **MYL2** | c.-52-15019A>G | chr12:110935600 | rs4426183 | Heterozygous | Tier2 | Not reviewed | 0.5 | — | 0.3 |
| **MYO18B** | c.6157-7087G>T | chr22:25985276 | rs9306420 | Homozygous | Tier2 | Not reviewed | 13.1 | — | 1.5 |
| **MYOT** | c.-233C>A | chr5:137867932 | rs186433387 | Heterozygous | Tier3 | Benign | 16.3 | — | 18.8 |
| **NDUFS2** | p.Arg323Gln | chr1:161210692 | rs35086265 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 16.8 | 0.525 | 8.7 |
| **NEB** | c.*201A>T | chr2:151485559 | rs190132249 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 1.9 | — | 3.7 |
| **NEB** | p.Ala4223Thr | chr2:151606686 | rs12998234 | Homozygous | Tier2 | Benign/Likely benign | 8.2 | 0.045 | 9.3 |
| **NEB** | p.Glu191Gln | chr2:151724301 | rs35686968 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 24.0 | 0.324 | 25.3 |
| **NF1** | c.*126G>A | chr17:31374281 | rs184715593 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 1.9 | — | 6.9 |
| **NFIX** | c.27+6825_27+6853del | chr19:13002688 | chr19:13002688 | Heterozygous | Tier2 | Not reviewed | — | — | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **NFIX** | `c.559+1309_559+1310del` | `chr19:13026858` | `rs1169659113` | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **NFIX** | `c.1255-191dup` | `chr19:13087799` | `chr19:13087799` | Homozygous | Tier2 | Not reviewed | 14.4 | — | — |
| **NIPBL** | `c.-79-1552_-79-1545del` | `chr5:36952054` | `chr5:36952054` | Heterozygous | Tier3 | Benign | — | — | — |
| **NOTCH1** | `p.Arg1279His` | `chr9:136506781` | `rs61751543` | Heterozygous | Tier3 | Benign/Likely benign | 22.5 | 0.098 | 14.1 |
| **NSUN2** | `p.Val715Ile` | `chr5:6600087` | `rs112951498` | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 2.0 | 0.011 | 11.0 |
| **PCSK9** | `c.207+15A>G` | `chr1:55040059` | `rs2495482` | Homozygous | Tier2 | Benign | 8.7 | — | 3.8 |
| **PCSK9** | `c.658-7C>T` | `chr1:55052643` | `rs2483205` | Homozygous | Tier2 | Benign | 1.3 | — | 0.3 |
| **PCSK9** | `c.799+3A>G` | `chr1:55052794` | `rs2495477` | Homozygous | Tier2 | Benign | 14.3 | — | 13.7 |
| **PCSK9** | `p.Gln342=` | `chr1:55057360` | `rs509504` | Homozygous | Tier2 | Conflicting classifications of pathogenicity | 8.6 | — | 13.3 |
| **PCSK9** | `p.Val460=` | `chr1:55058524` | `rs540796` | Heterozygous | Tier2 | Benign | 8.8 | — | 15.2 |
| **PCSK9** | `p.Val474Ile` | `chr1:55058564` | `rs562556` | Heterozygous | Tier2 | Benign/Likely benign | 0.0 | 0.051 | 0.2 |
| **PCSK9** | `p.Gly670Glu` | `chr1:55063514` | `rs505151` | Homozygous | Tier2 | Benign/Likely benign | 1.7 | 0.067 | 12.1 |
| **PCSK9** | `c.*571C>T` | `chr1:55064155` | `rs662145` | Heterozygous | Tier2 | Benign | 0.7 | — | 0.3 |
| **PEX10** | `p.Thr274Ala` | `chr1:2406576` | `rs34154371` | Heterozygous | Tier3 | Benign/Likely benign | 23.6 | 0.357 | 18.0 |
| **PEX13** | `p.Leu360Val` | `chr2:61048636` | `rs74350038` | Heterozygous | Tier3 | Benign/Likely benign | 1.8 | 0.151 | 11.0 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **PEX16** | p.Val254Leu | chr11:45914138 | rs35214605 | Heterozygous | Tier3 | Benign/Likely benign | 15.4 | 0.046 | 18.8 |
| **PIBF1** | p.His523Asn | chr13:72908609 | rs115858628 | Heterozygous | Tier3 | Benign/Likely benign | 19.6 | 0.085 | 19.7 |
| **PKP2** | p.Val543Ile | chr12:32824092 | rs146102241 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 23.2 | 0.333 | 22.9 |
| **PKP2** | p.Ser70Ile | chr12:32896523 | rs75909145 | Heterozygous | Tier3 | Benign/Likely benign | 21.6 | 0.155 | 14.8 |
| **PLEC** | p.Arg296Gln | chr8:143934868 | rs138924815 | Heterozygous | Tier3 | Benign | 24.6 | 0.398 | 22.0 |
| **PNPLA2** | p.Pro265Ser | chr11:823729 | rs56152088 | Heterozygous | Tier3 | Benign | 13.9 | 0.111 | 19.6 |
| **POGZ** | c.284-4376T>C | chr1:151435217 | rs11578888 | Homozygous | Tier2 | Not reviewed | 0.1 | — | 0.5 |
| **POLR3A** | c.*620G>A | chr10:77976858 | rs184043149 | Heterozygous | Tier2 | Uncertain significance | 0.0 | — | 0.0 |
| **POMT1** | p.Asp411Glu | chr9:131515483 | rs11243406 | Heterozygous | Tier3 | Benign | 23.9 | 0.464 | 24.3 |
| **RASA1** | p.Ala99Val | chr5:87268747 | rs111840875 | Heterozygous | Tier3 | Benign/Likely benign | 15.9 | 0.189 | 16.6 |
| **RYR1** | p.Leu198= | chr19:38444640 | rs2229139 | Heterozygous | Tier2 | Benign/Likely benign | 10.4 | — | 12.1 |
| **RYR1** | p.Ala359= | chr19:38448768 | rs10406027 | Homozygous | Tier2 | Benign | 0.4 | — | 0.0 |
| **RYR1** | p.Asn993= | chr19:38466199 | rs2228070 | Heterozygous | Tier2 | Benign | 0.9 | — | 0.0 |
| **RYR1** | p.Ile1152= | chr19:38469040 | rs11083462 | Heterozygous | Tier2 | Benign | 5.6 | — | 0.0 |
| **RYR1** | p.Ala2500= | chr19:38500876 | rs2228072 | Heterozygous | Tier2 | Benign | 0.8 | — | 0.0 |
| **RYR1** | p.Val2509= | chr19:38500903 | rs2071088 | Heterozygous | Tier2 | Benign | 4.2 | — | 11.0 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **RYR1** | c.7835+5A>G | chr19:38502732 | rs1469695 | Heterozygous | Tier2 | Benign | 2.2 | — | 9.1 |
| **RYR1** | p.His2621= | chr19:38502907 | rs2229142 | Heterozygous | Tier2 | Benign | 6.4 | — | 11.9 |
| **RYR1** | p.Thr2659= | chr19:38504270 | rs2229144 | Heterozygous | Tier2 | Benign | 0.5 | — | 0.0 |
| **RYR1** | p.Ile2706= | chr19:38504798 | rs2960340 | Heterozygous | Tier2 | Benign | 0.8 | — | 0.2 |
| **RYR1** | p.Asp2730= | chr19:38504870 | rs2915951 | Heterozygous | Tier2 | Benign | 0.3 | — | 0.0 |
| **RYR1** | p.Glu2779= | chr19:38505335 | rs2915952 | Heterozygous | Tier2 | Benign | 6.4 | — | 12.9 |
| **RYR1** | p.Ser2863= | chr19:38506350 | rs2229146 | Heterozygous | Tier2 | Benign | 3.7 | — | 0.1 |
| **RYR1** | c.8693-10G>C | chr19:38506819 | rs2915958 | Heterozygous | Tier2 | Benign | 5.2 | — | 4.1 |
| **RYR1** | p.Pro3062= | chr19:38512085 | rs2071089 | Heterozygous | Tier2 | Benign | 0.1 | — | 0.0 |
| **RYR2** | p.Ser1400Gly | chr1:237591776 | rs56229512 | Heterozygous | Tier3 | Benign/Likely benign | 19.2 | 0.310 | 17.5 |
| **SCAPER** | c.1866+8A>G | chr15:76753800 | rs139187840 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 2.4 | — | 7.1 |
| **SCN4A** | c.*1069G>A | chr17:63939702 | rs368616841 | Heterozygous | Tier2 | Uncertain significance | 0.0 | — | 0.0 |
| **SGCD** | c.-44+62149G>C | chr5:156186168 | rs564618852 | Heterozygous | Tier2 | Not reviewed | 6.7 | — | 2.6 |
| **SIX3** | c.-1863G>A | chr2:44940242 | rs1353919432 | Heterozygous | Tier2 | Not reviewed | 20.2 | — | 19.3 |
| **SMARCA4** | c.4425-5369C>T | chr19:11052886 | rs1529729 | Heterozygous | Tier2 | Not reviewed | 0.6 | — | 1.7 |
| **SMARCA4** | c.*327T>C | chr19:11062143 | rs146968124 | Heterozygous | Tier3 | Benign | 15.2 | — | 12.5 |
| **SMO** | p.Pro698Arg | chr7:129212180 | rs116640950 | Heterozygous | Tier2 | Likely benign | 12.4 | 0.153 | 19.9 |
| **SMPX** | c.*14+6246G>C | chrX:21731303 | rs1015473 | Heterozygous | Tier2 | Not reviewed | 1.1 | — | 9.4 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **SON** | p.Tyr1230_Thr1247dup | chr21:33552918 | rs1569055881 | Heterozygous | Tier3 | Likely benign | — | — | — |
| **SPEG** | p.Pro674Arg | chr2:219449179 | rs757668535 | Heterozygous | Tier2 | Uncertain significance | 12.8 | 0.050 | 20.7 |
| **SPEN** | p.Glu1517Val | chr1:15930790 | rs143050516 | Heterozygous | Tier2 | Likely benign | 25.3 | 0.109 | 22.7 |
| **SPTBN1** | c.-48+1068G>T | chr2:54457586 | rs995589477 | Heterozygous | Tier2 | Not reviewed | 20.5 | — | 20.4 |
| **STIL** | c.*545C>T | chr1:47250591 | rs181264069 | Heterozygous | Tier2 | Uncertain significance | 1.4 | — | 0.7 |
| **STRA6** | p.Arg408Gln | chr15:74183933 | rs150814749 | Heterozygous | Tier2 | Benign/Likely benign | 0.6 | 0.159 | 16.5 |
| **SYNE2** | c.18038+4450A>G | chr14:64194687 | rs748933084 | Heterozygous | Tier2 | Not reviewed | 9.4 | — | 9.0 |
| **TARS1** | c.330-4del | chr5:33453265 | chr5:33453265 | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **THSD4** | c.1153-113066G>A | chr15:71547464 | rs117286812 | Heterozygous | Tier3 | Benign | 18.2 | 0.049 | 23.8 |
| **TMEM107** | c.*622G>A | chr17:8173581 | rs201266955 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 14.5 | — | 22.9 |
| **TMEM43** | p.Trp316Ser | chr3:14139244 | rs199526104 | Heterozygous | **Tier1** | Conflicting classifications of pathogenicity | 32.0 | 0.533 | 26.5 |
| **TMEM43** | c.*115T>C | chr3:14141910 | rs1043943 | Heterozygous | Tier2 | Benign/Likely benign | 1.6 | — | 9.2 |
| **TRIP11** | p.Arg1576His | chr14:91999405 | rs35007347 | Heterozygous | Tier3 | Benign/Likely benign | 23.4 | 0.137 | 22.0 |
| **TTN** | p.Asn16662Thr | chr2:178612540 | rs36043230 | Heterozygous | Tier3 | Benign/Likely benign | 23.3 | 0.328 | 22.0 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) |
|---|---|---|---|---|---|---|---|---|---|
| **TTN** | `p.Pro10462Thr` | `chr2:178694641` | `rs761102022` | Heterozygous | Tier2 | Uncertain significance | 20.1 | 0.381 | 22.6 |
| **TTR** | `c.-1054C>G` | `chr18:31556179` | `chr18:31556179` | Heterozygous | Tier2 | Not reviewed | 8.6 | — | 15.1 |
| **VCL** | `p.His27=` | `chr10:73998288` | `rs200733607` | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 13.7 | — | 16.0 |
| **VCL** | `c.3153+47T>C` | `chr10:74114434` | `rs2270550` | Homozygous | Tier2 | Benign | 0.3 | — | 5.1 |
| **VPS13A** | `p.Gln2640=` | `chr9:77357805` | `rs200387635` | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 5.4 | — | 0.3 |
| **WAC** | `c.274+13301C>G` | `chr10:28549058` | `rs913821` | Heterozygous | Tier2 | Not reviewed | 0.1 | — | 0.5 |
| **WARS2** | `p.His151Arg` | `chr1:119042327` | `rs150022801` | Heterozygous | Tier2 | Likely benign | 14.3 | 0.062 | 17.8 |
| **WFS1** | `c.-5-2548G>C` | `chr4:6274903` | `rs62283056` | Heterozygous | Tier2 | Not reviewed | 3.4 | — | 7.9 |
| **WRN** | `c.2967+88G>A` | `chr8:31132594` | `rs2252480` | Heterozygous | Tier2 | Benign | 1.3 | — | 0.7 |
| **ZEB2** | `c.74-11140_74-11139dup` | `chr2:144441167` | `chr2:144441167` | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **ZEB2** | `c.-69-197A>T` | `chr2:144517616` | `rs533595036` | Heterozygous | Tier2 | Not reviewed | 20.7 | — | 18.6 |
| **ZEB2** | `c.-69-198C>G` | `chr2:144517617` | `rs558617021` | Heterozygous | Tier2 | Not reviewed | 17.8 | — | 15.7 |
| **ZFPM2** | `c.420+34535_420+34536del` | `chr8:105596005` | `chr8:105596005` | Heterozygous | Tier2 | Not reviewed | — | — | — |
| **ZFPM2** | `c.533-57576C>T` | `chr8:105731142` | `rs544662446` | Heterozygous | Tier2 | Not reviewed | 19.7 | — | 14.7 |

## 4. Metabolic, Mitochondrial & DNA Repair Machinery

0 Variants

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | External Accessions |
|---|---|---|---|---|---|---|---|---|---|---|

## 5. Protective & Longevity Modifiers

9 Variants

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL | AVI (Phred) | Extern Acces |
|---|---|---|---|---|---|---|---|---|---|---|
| **ADH1C** | p.Ile350Val | chr4:99339632 | rs698 | Heterozygous | Tier2 | protective | 15.4 | — | 15.9 | VCV18 |
| **ADH1C** | p.Arg272Gln | chr4:99342808 | rs1693482 | Heterozygous | Tier2 | protective | 10.4 | — | 12.8 | VCV18 |
| **CASP8** | c.-937_-932del | chr2:201232809 | rs3834129 | Heterozygous | Tier3 | protective | — | — | — | VCV77 |
| **CD209** | c.-336T>C | chr19:7747847 | rs4804803 | Heterozygous | Tier3 | protective\|risk factor | 3.8 | — | 1.9 | VCV54 |
| **EDNRA** | c.-230G>A | chr4:147481217 | rs1801708 | Heterozygous | Tier3 | protective | 16.2 | — | 13.5 | VCV16 |
| **FCGR2B** | p.Ile232Thr | chr1:161674008 | rs1050501 | Homozygous | Tier3 | protective\|risk factor | 10.9 | 0.024 | 11.2 | VCV54 |
| **GNPTAB** | c.117+9523G>T | chr12:101821036 | rs11111044 | Heterozygous | Tier2 | Not reviewed | 0.2 | — | 0.4 | — / At |
| **GNPTAB** | c.117+3554G>C | chr12:101827005 | rs7964859 | Heterozygous | Tier2 | Not reviewed | 0.7 | — | 0.6 | — / At |
| **TLR1** | p.Ser602Ile | chr4:38797027 | rs5743618 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity\|protective | 7.5 | 0.056 | 7.9 | VCV83 |

## 6. Secondary & Exploratory Clinical Variants

556 Variants

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **ABCA12** | p.Lys982Thr | chr2:215000939 | rs142196906 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 18.4 | 0.354 |
| **ABCC4** | c.3871-6349C>T | chr13:95028031 | rs4148546 | Heterozygous | Tier2 | Not reviewed | 1.9 | — |
| **ABCD3** | p.Ala131Thr | chr1:94473821 | rs138378383 | Heterozygous | Tier2 | Uncertain significance | 22.9 | 0.492 |
| **ACAA2** | p.Val253Ile | chr18:49791596 | rs377532103 | Heterozygous | Tier2 | Likely benign | 1.4 | 0.307 |
| **ACADSB** | c.*3332A>T | chr10:123057097 | rs763405920 | Heterozygous | Tier2 | Uncertain significance | 10.4 | — |
| **ACHE** | p.His353Asn | chr7:100893176 | rs1799805 | Heterozygous | Tier3 | Benign | 19.0 | 0.038 |
| **ACOX3** | c.-15+600A>G | chr4:8440048 | rs62287361 | Heterozygous | **Tier1** | Not reviewed | 3.3 | — |
| **ACSF3** | p.Arg39Cys | chr16:89100796 | rs779775679 | Heterozygous | Tier2 | Uncertain significance | 4.0 | 0.038 |
| **ACSM2B** | p.Tyr540Asn | chr16:20540665 | rs139253186 | Heterozygous | **Tier1** | Not reviewed | 27.6 | 0.401 |
| **ACTN4** | p.Asp581Asn | chr19:38724205 | rs768314562 | Heterozygous | Tier2 | Not reviewed | 21.6 | 0.179 |
| **ACTR5** | p.Met148Ile | chr20:38750078 | rs34862565 | Heterozygous | Tier3 | Not reviewed | 23.7 | 0.757 |
| **ADAMTS18** | p.Pro12Arg | chr16:77434661 | rs200952997 | Heterozygous | Tier2 | Benign/Likely benign | 15.9 | 0.033 |
| **ADAMTS7** | c.1029-221G>A | chr15:78790059 | rs142754252 | Heterozygous | **Tier1** | Not reviewed | 0.2 | — |
| **ADAMTSL4** | p.Pro381Arg | chr1:150554375 | rs145639662 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 22.4 | 0.037 |
| **ADCY10** | p.Pro1353Leu | chr1:167823118 | rs72697797 | Heterozygous | Tier3 | Benign/Likely benign | 0.7 | 0.128 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **ADCY8** | c.1111-10720C>T | chr8:130962718 | rs36109123 | Heterozygous | Tier2 | Not reviewed | 4.9 | — |
| **ADGRV1** | p.Ala2803Thr | chr5:90705420 | rs111033530 | Heterozygous | Tier3 | Benign/Likely benign | 25.6 | 0.215 |
| **ADGRV1** | p.Thr4090Asn | chr5:90763453 | rs199839743 | Heterozygous | Tier2 | Benign/Likely benign | 22.6 | 0.211 |
| **AEBP1** | p.Ala1031Thr | chr7:44113875 | rs146344486 | Heterozygous | Tier2 | Benign/Likely benign | 23.2 | 0.258 |
| **AEBP1** | p.Val1148Ile | chr7:44114226 | rs13898 | Heterozygous | Tier3 | Benign/Likely benign | 20.9 | 0.221 |
| **AFG2B** | c.2094+15del | chr15:45419998 | chr15:45419998 | Heterozygous | Tier2 | Not reviewed | — | — |
| **AGT** | p.Ala286Ser | chr1:230706174 | rs377397992 | Heterozygous | Tier2 | Uncertain significance | 1.3 | 0.133 |
| **AGTR1** | c.-47-8809C>T | chr3:148732180 | rs146439726 | Heterozygous | Tier2 | Not reviewed | 7.8 | — |
| **AIFM3** | p.Arg401Trp | chr22:20976921 | rs1365055212 | Heterozygous | Tier2 | Not reviewed | 29.2 | 0.277 |
| **AIMP1** | p.Glu120Gly | chr4:106328211 | rs367813302 | Heterozygous | Tier2 | Uncertain significance | 20.3 | 0.100 |
| **AIRE** | c.1279-18C>T | chr21:44293771 | rs72650678 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 2.8 | — |
| **AK4P3** | p.Ala177Val | chr12:31615910 | rs35037649 | Homozygous | Tier2 | Not reviewed | 7.1 | — |
| **AKAP1** | p.Ser330Gly | chr17:57106452 | rs34535433 | Heterozygous | Tier3 | Benign | 0.1 | 0.021 |
| **AKAP5** | p.Ser205Phe | chr14:64469008 | rs2230492 | Heterozygous | Tier2 | Not reviewed | 23.1 | 0.176 |
| **ALOXE3** | c.*514T>G | chr17:8096113 | rs80223722 | Heterozygous | Tier2 | Uncertain significance | 13.1 | — |
| **ALPK1** | c.-101+27968G>A | chr4:112343820 | chr4:112343820 | Heterozygous | Tier2 | Not reviewed | 17.2 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **ALPL** | p.Leu252= | chr1:21568211 | rs867145238 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 9.6 | — |
| **AMPD1** | p.Glu435Gly | chr1:114677435 | rs764950035 | Heterozygous | **Tier1** | Uncertain significance | 32.0 | 0.949 |
| **ANK2** | c.85-22668G>A | chr4:113151748 | rs11929834 | Heterozygous | Tier2 | Not reviewed | 8.0 | — |
| **ANO1** | p.Arg147fs | chr11:70089999 | chr11:70089999 | Heterozygous | Tier2 | Not reviewed | — | — |
| **APOL1** | p.Leu7Val | chr22:36254974 | rs372661836 | Heterozygous | Tier2 | Uncertain significance | 14.6 | 0.044 |
| **APOO** | c.389-16A>G | chrX:23858749 | rs373181632 | Heterozygous | Tier2 | Not reviewed | 14.2 | — |
| **AR** | p.Gln67_Gln80del | chrX:67545317 | chrX:67545317 | Heterozygous | Tier3 | Likely benign | — | — |
| **AR** | p.Gly461_Gly473del | chrX:67546515 | chrX:67546515 | Heterozygous | Tier3 | Benign | — | — |
| **ARG1** | c.306-388A>G | chr6:131580831 | rs779800455 | Heterozygous | Tier2 | Not reviewed | 7.6 | — |
| **ARHGEF2** | p.Ser20ProfsTer? | chr1:156006982 | rs558141032 | Heterozygous | Tier2 | Not reviewed | — | — |
| **ASAH2** | p.Ala623Ser | chr10:50196910 | rs1444286757 | Heterozygous | Tier2 | Not reviewed | 19.1 | 0.108 |
| **ASB3** | c.1238+1126G>A | chr2:53699145 | rs72793655 | Heterozygous | Tier2 | Not reviewed | 0.9 | — |
| **ASXL3** | c.54+3289C>T | chr18:33581974 | rs538726349 | Heterozygous | Tier2 | Not reviewed | 18.7 | — |
| **ATG4B** | p.Pro188Ala | chr2:241666668 | rs201601658 | Heterozygous | Tier2 | Not reviewed | 8.8 | 0.023 |
| **ATP1A1** | p.Ile580Val | chr1:116395187 | rs77217304 | Heterozygous | Tier3 | Benign | 1.1 | 0.053 |
| **ATP6V0A2** | p.Asn850Ser | chr12:123758010 | rs75279884 | Heterozygous | Tier3 | Benign | 0.0 | 0.191 |
| **ATP6V0A4** | c.-29G>A | chr7:138786169 | rs145258624 | Heterozygous | Tier2 | Uncertain significance | 7.2 | — |
| **ATP8A2** | c.2384+2419_2384+2422dup | chr13:25701726 | chr13:25701726 | Heterozygous | Tier2 | Not reviewed | — | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **ATR** | p.Val959Met | chr3:142550233 | rs28910271 | Heterozygous | Tier3 | Benign | 12.7 | 0.065 |
| **ATR** | p.Val316Ile | chr3:142562456 | rs28897764 | Heterozygous | Tier3 | Benign | 17.3 | 0.034 |
| **ATXN1L** | p.Pro597His | chr16:71851530 | rs61747556 | Heterozygous | Tier2 | Not reviewed | 19.3 | 0.058 |
| **ATXN2** | p.Gln25_Gln28dup | chr12:111598964 | rs1555246688 | Heterozygous | Tier3 | Benign | — | — |
| **AWAT2** | p.Ile96Val | chrX:70043664 | rs61735680 | Heterozygous | Tier2 | Not reviewed | 16.2 | 0.126 |
| **BACH2** | c.-12-16333C>T | chr6:90025189 | rs4388268 | Heterozygous | Tier2 | Not reviewed | 4.6 | — |
| **BCAS3** | c.1487-5702C>T | chr17:61010049 | rs571142153 | Heterozygous | Tier2 | Not reviewed | 17.4 | — |
| **BCKDHB** | c.951+20710T>G | chr6:80223922 | chr6:80223922 | Heterozygous | Tier2 | Not reviewed | 16.6 | — |
| **BCL11A** | c.386-8965T>A | chr2:60477798 | rs7593947 | Homozygous | Tier2 | Not reviewed | 2.0 | — |
| **BCL11B** | c.58+3615_58+3616dup | chr14:99267548 | chr14:99267548 | Heterozygous | Tier2 | Not reviewed | — | — |
| **BLTP1** | p.Pro802Ala | chr4:122219496 | rs1048893554 | Heterozygous | Tier2 | Uncertain significance | 23.7 | 0.409 |
| **BMP8B** | c.674-430G>C | chr1:39765247 | rs183659449 | Heterozygous | Tier2 | Not reviewed | 6.1 | — |
| **BMP8B** | p.Gly92Cys | chr1:39788212 | rs769660218 | Heterozygous | Tier2 | Uncertain significance | 22.7 | 0.159 |
| **BNC2** | c.3+1897C>T | chr9:16868749 | rs1053680531 | Heterozygous | Tier2 | Not reviewed | 16.7 | — |
| **BPTF** | c.1865-5726A>G | chr17:67886118 | rs117415449 | Heterozygous | Tier3 | Benign | 21.7 | — |
| **BSN** | p.Ser3307Gly | chr3:49661764 | rs145931220 | Heterozygous | Tier2 | Likely benign | 1.8 | 0.075 |
| **C10orf67** | p.Ala524fs | chr10:23209999 | chr10:23209999 | Heterozygous | Tier2 | Not reviewed | — | — |
| **C10orf67** | c.1571-5745_1571-5744del | chr10:23210001 | chr10:23210001 | Heterozygous | Tier2 | Not reviewed | — | — |
| **C4A** | p.Ser347Tyr | chr6:31984403 | rs392610 | Heterozygous | Tier2 | Benign | 26.5 | 0.212 |
| **C4A** | p.Arg477Trp | chr6:31991819 | rs1554297700 | Heterozygous | Tier2 | Not reviewed | 15.9 | 0.183 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **C4B** | p.Thr907Ala | chr6:32027362 | rs796750528 | Heterozygous | Tier2 | Benign/Likely benign | 5.9 | 0.034 |
| **CA8** | c.418-13115A>G | chr8:60245494 | rs9643367 | Heterozygous | Tier2 | Not reviewed | 16.8 | — |
| **CACNA1E** | c.3829-920G>A | chr1:181754317 | rs147545823 | Homozygous | Tier2 | Not reviewed | 10.6 | — |
| **CACNA1F** | p.Arg508Gln | chrX:49226037 | rs34162630 | Heterozygous | Tier3 | Benign/Likely benign | 25.9 | 0.715 |
| **CACNA1G** | c.354+31_354+34del | chr17:50568986 | chr17:50568986 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | — | — |
| **CACNA1H** | c.1120-3C>T | chr16:1200713 | rs559931034 | Heterozygous | Tier2 | Uncertain significance | 5.8 | — |
| **CACNA1H** | p.Arg2218His | chr16:1220585 | rs56885166 | Heterozygous | Tier2 | Benign/Likely benign | 25.6 | 0.282 |
| **CACNA2D1** | c.294+36117T>C | chr7:82299018 | rs6969265 | Heterozygous | Tier2 | Not reviewed | 1.1 | — |
| **CANT1** | p.Arg19Gln | chr17:78997567 | rs144060377 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 24.5 | 0.559 |
| **CAPN2** | p.Glu521Gln | chr1:223761612 | rs28370127 | Heterozygous | Tier2 | Not reviewed | 21.8 | 0.219 |
| **CARD10** | p.Arg601Gln | chr22:37497164 | rs201423754 | Heterozygous | Tier2 | Uncertain significance | 10.7 | 0.070 |
| **CARMIL2** | p.Lys937del | chr16:67652329 | rs144268948 | Heterozygous | Tier3 | Benign | — | — |
| **CARS2** | p.Gly216Glu | chr13:110683059 | rs146773721 | Heterozygous | Tier3 | Benign | 1.5 | 0.060 |
| **CATSPER2** | c.-159+765G>A | chr15:43663653 | rs1555451673 | Heterozygous | Tier3 | Not reviewed | 18.4 | — |
| **CATSPER2** | c.-248T>C | chr15:43664507 | rs1306890410 | Heterozygous | Tier2 | Not reviewed | 21.1 | — |
| **CATSPER2** | c.-285+323G>A | chr15:43667705 | rs143039626 | Homozygous | Tier2 | Not reviewed | 20.3 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **CCDC80** | p.Tyr938His | chr3:112605458 | rs114697626 | Heterozygous | Tier2 | Likely benign | 26.5 | 0.235 |
| **CD109** | p.Lys789Glu | chr6:73787261 | rs45572237 | Heterozygous | Tier2 | Not reviewed | 8.5 | 0.045 |
| **CD177** | c.620-1440T>G | chr19:43358825 | rs1568444200 | Homozygous | Tier2 | Not reviewed | 2.2 | — |
| **CD177** | c.620-1117A>G | chr19:43359148 | rs71339505 | Homozygous | Tier2 | Not reviewed | 6.3 | — |
| **CD177** | c.620-1074T>C | chr19:43359191 | rs71339506 | Homozygous | Tier2 | Not reviewed | 7.5 | — |
| **CD177** | c.620-1022T>C | chr19:43359243 | rs755694960 | Homozygous | Tier2 | Not reviewed | 2.2 | — |
| **CD40** | c.559+36G>C | chr20:46126737 | rs41282788 | Heterozygous | **Tier1** | Benign | 2.0 | — |
| **CDAN1** | p.Val328Ile | chr15:42735336 | rs144448301 | Heterozygous | Tier2 | Uncertain significance | 17.1 | 0.237 |
| **CDH1** | c.1711+343C>T | chr16:68819768 | rs4783689 | Homozygous | Tier2 | Not reviewed | 0.6 | — |
| **CDH12** | c.527-17594G>A | chr5:21872384 | rs1032542229 | Heterozygous | Tier2 | Not reviewed | 17.1 | — |
| **CDH8** | c.1537-6038_1537-6037del | chr16:61719997 | chr16:61719997 | Heterozygous | Tier2 | Not reviewed | — | — |
| **CDHR1** | p.His53Gln | chr10:84196512 | rs12781048 | Heterozygous | Tier3 | Benign | 0.0 | 0.144 |
| **CEBPE** | p.Leu155Met | chr14:23118629 | rs141903485 | Heterozygous | Tier2 | Benign/Likely benign | 25.6 | 0.113 |
| **CELA1** | p.Tyr5Ser | chr12:51346625 | rs117443541 | Homozygous | Tier2 | Not reviewed | 14.8 | 0.171 |
| **CELA1** | p.Tyr5His | chr12:51346626 | rs116944010 | Homozygous | Tier2 | Not reviewed | 15.2 | 0.206 |
| **CELA1** | p.Leu4AlafsTer21 | chr12:51346630 | rs377599213 | Homozygous | Tier2 | Benign | 0.8 | — |
| **CELA1** | p.Val3ProfsTer21 | chr12:51346631 | rs1491233894 | Homozygous | Tier2 | Benign | — | — |
| **CEP63** | p.Gln185His | chr3:134537268 | rs114108011 | Heterozygous | **Tier1** | Benign | 34.0 | 0.276 |
| **CEP85L** | c.1020+5170C>T | chr6:118560359 | rs537800022 | Heterozygous | Tier2 | Uncertain significance | 0.2 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **CERS2** | p.Val267Ile | chr1:150966805 | rs149992226 | Heterozygous | Tier2 | Likely benign | 0.1 | 0.259 |
| **CES1** | p.Arg4Pro | chr16:55833045 | rs111604615 | Heterozygous | Tier2 | Not reviewed | 8.5 | 0.145 |
| **CHD2** | c.1153+109G>C | chr15:92944624 | rs1320819 | Heterozygous | Tier2 | Benign | 0.2 | — |
| **CHD2** | c.1502+505G>C | chr15:92949581 | rs4327000 | Heterozygous | Tier2 | Not reviewed | 1.4 | — |
| **CHODL** | c.-45+109054_-45+109055insGAA | chr21:18026454 | chr21:18026454 | Homozygous | Tier2 | Not reviewed | — | — |
| **CHRD** | p.Pro10Ser | chr3:184380346 | chr3:184380346 | Heterozygous | Tier2 | Not reviewed | 21.7 | 0.015 |
| **CHRNE** | p.Gly18Val | chr17:4902757 | rs4790235 | Heterozygous | Tier3 | Benign | 7.0 | 0.128 |
| **CLEC4M** | p.Arg187Gln | chr19:7765983 | rs28413926 | Homozygous | Tier2 | Not reviewed | 0.0 | 0.027 |
| **CLEC4M** | p.Glu210ArgfsTer3 | chr19:7766049 | rs2034253170 | Homozygous | Tier2 | Not reviewed | — | — |
| **CLEC4M** | p.Glu210AlafsTer7 | chr19:7766051 | rs867796501 | Homozygous | Tier2 | Not reviewed | — | — |
| **CNGB1** | c.159+14C>T | chr16:57967114 | rs199591689 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 3.0 | — |
| **CNTN6** | c.762-5763G>A | chr3:1315887 | rs155231 | Homozygous | Tier2 | Not reviewed | 0.2 | — |
| **CNTNAP2** | c.*3999del | chr7:148419601 | chr7:148419601 | Heterozygous | Tier2 | Uncertain significance | — | — |
| **CNTNAP3C** | c.85+1G>A | chr9:61331053 | rs71506495 | Heterozygous | Tier2 | Not reviewed | 25.8 | — |
| **CNTNAP5** | c.2077+37493_2077+37494insA | chr2:124685452 | rs201059908 | Heterozygous | Tier2 | Not reviewed | 20.2 | — |
| **CNTNAP5** | c.2077+37494T>A | chr2:124685452 | rs149834795 | Heterozygous | Tier2 | Not reviewed | 20.2 | — |
| **CNTNAP5** | c.2077+37495T>A | chr2:124685453 | rs142344379 | Heterozygous | Tier2 | Not reviewed | 21.6 | — |
| **CNTNAP5** | p.Pro715Arg | chr2:124747295 | rs200759902 | Heterozygous | Tier2 | Not reviewed | 13.9 | 0.018 |
| **COG1** | p.Ala20Pro | chr17:73193127 | rs142719529 | Heterozygous | Tier3 | Benign/Likely benign | 29.8 | 0.168 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **COG5** | c.1476-14T>G | chr7:107281413 | rs769339809 | Heterozygous | Tier3 | Likely benign | 17.0 | — |
| **COG5** | c.538+54830T>C | chr7:107472407 | chr7:107472407 | Heterozygous | Tier2 | Not reviewed | 18.9 | — |
| **COL15A1** | p.Phe851Leu | chr9:99042084 | rs35901514 | Heterozygous | Tier3 | Benign | 23.9 | 0.412 |
| **COL23A1** | c.361+42073T>G | chr5:178518609 | rs1350094006 | Heterozygous | Tier2 | Not reviewed | 7.6 | — |
| **COL28A1** | p.Gly1066Glu | chr7:7360398 | rs200146803 | Heterozygous | Tier2 | Uncertain significance | 1.4 | 0.110 |
| **COL4A1** | c.1286-281T>C | chr13:110195399 | rs12184772 | Heterozygous | Tier3 | Benign | 0.0 | — |
| **COL4A2** | c.4882-83T>C | chr13:110511851 | rs2479426 | Heterozygous | Tier3 | Benign | 1.5 | — |
| **COL4A4** | c.72-854C>T | chr2:227145412 | rs2001851 | Heterozygous | Tier2 | Not reviewed | 0.5 | — |
| **CPS1** | c.3+12249del | chr2:210489999 | chr2:210489999 | Heterozygous | Tier2 | Not reviewed | — | — |
| **CPSF3** | p.Lys473Arg | chr2:9452935 | rs1128215 | Heterozygous | Tier2 | Likely benign | 20.2 | 0.065 |
| **CR1** | p.Thr895Ala | chr1:207550815 | rs147353507 | Homozygous | Tier2 | Not reviewed | 6.1 | 0.063 |
| **CRACR2A** | c.1602+6G>A | chr12:3638118 | rs145536162 | Heterozygous | Tier3 | Likely benign | 2.5 | — |
| **CSF3R** | p.Glu808Lys | chr1:36466446 | rs146617729 | Heterozygous | Tier2 | Benign/Likely benign | 23.2 | 0.086 |
| **CSF3R** | c.1865-65G>A | chr1:36467716 | rs3917999 | Heterozygous | Tier2 | Benign | 2.9 | — |
| **CTH** | p.Thr67Ile | chr1:70415987 | rs28941785 | Heterozygous | **Tier1** | Conflicting classifications of pathogenicity | 25.5 | 0.931 |
| **CTU2** | p.Thr424Ile | chr16:88714656 | rs373776054 | Heterozygous | Tier2 | Uncertain significance | 10.4 | 0.128 |
| **CUBN** | p.Pro1975Leu | chr10:16937594 | rs41289303 | Heterozygous | Tier3 | Benign | 25.3 | 0.341 |
| **CUX2** | c.301+6815A>C | chr12:111270654 | rs1265564 | Heterozygous | Tier2 | Not reviewed | 0.7 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **CYP1A1** | p.Thr461Asn | chr15:74720646 | rs1799814 | Heterozygous | Tier3 | Benign | 0.6 | 0.118 |
| **CYP21A2** | p.Ser494Asn | chr6:32041127 | rs6473 | Heterozygous | Tier2 | Benign | 0.2 | 0.076 |
| **CYP2D6** | p.Lys281del | chr22:42128174 | rs5030656 | Heterozygous | Tier3 | Likely benign\|other | — | — |
| **CYP4A22** | p.Leu72Ile | chr1:47140798 | rs56258513 | Heterozygous | Tier2 | Not reviewed | 15.4 | 0.135 |
| **DBH** | p.Asn201Ser | chr9:133642322 | rs45465204 | Heterozygous | Tier3 | Benign | 10.7 | 0.184 |
| **DCAF17** | c.*1841G>T | chr2:171482955 | rs114519296 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 4.3 | — |
| **DCHS2** | p.Glu3241Asp | chr4:154234929 | rs753533849 | Heterozygous | Tier2 | Not reviewed | 20.5 | 0.280 |
| **DCPS** | p.Asp74Gly | chr11:126306589 | rs145173937 | Heterozygous | Tier2 | Not reviewed | 20.8 | 0.036 |
| **DDR2** | c.-1310T>C | chr1:162630329 | rs7544972 | Heterozygous | Tier2 | Not reviewed | 0.4 | — |
| **DEFA5** | p.Arg71His | chr8:7055504 | rs7839771 | Heterozygous | Tier3 | Benign | 0.0 | 0.049 |
| **DHFR** | c.*1874C>T | chr5:80627213 | rs918868210 | Heterozygous | Tier2 | Not reviewed | 3.8 | — |
| **DMAP1** | p.Glu239Gly | chr1:44218751 | chr1:44218751 | Heterozygous | **Tier1** | Not reviewed | 29.7 | 0.377 |
| **DNAH3** | c.5884+7486C>G | chr16:21012138 | rs2087632048 | Heterozygous | Tier2 | Not reviewed | 8.3 | — |
| **DOCK2** | p.Ala1816Ser | chr5:170082811 | rs144315682 | Heterozygous | Tier2 | Benign/Likely benign | 0.3 | 0.016 |
| **DSC1** | p.Gly740Arg | chr18:31132588 | rs147041330 | Heterozygous | Tier2 | Not reviewed | 25.8 | 0.287 |
| **DTNBP1** | p.Pro272Ser | chr6:15523217 | rs17470454 | Heterozygous | Tier3 | Benign/Likely benign | 21.3 | 0.134 |
| **DYNLL2** | c.*1936A>G | chr17:58091215 | rs539751637 | Heterozygous | Tier2 | Not reviewed | 18.6 | — |
| **EIF2AK1** | p.Leu319His | chr7:6041055 | rs34909691 | Heterozygous | Tier3 | Benign | 7.4 | 0.028 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **ELOVL4** | p.Glu272Gln | chr6:79916739 | rs148919174 | Heterozygous | Tier3 | Benign/Likely benign | 10.9 | 0.016 |
| **EMC1** | c.96-708A>G | chr1:19245738 | rs61764877 | Homozygous | Tier2 | Not reviewed | 1.8 | — |
| **EML1** | c.67+28559T>G | chr14:99822102 | rs117232270 | Heterozygous | Tier2 | Not reviewed | 4.4 | — |
| **EMSY** | c.877-501T>A | chr11:76472063 | rs368463439 | Heterozygous | Tier2 | Not reviewed | 13.0 | — |
| **EPHA2** | p.Arg721Gln | chr1:16132227 | rs116506614 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 28.2 | 0.460 |
| **ERICH3** | p.Asp1463His | chr1:74571323 | rs12131196 | Heterozygous | Tier2 | Not reviewed | 6.0 | 0.030 |
| **ETNK2** | c.518+1064C>T | chr1:204148639 | rs78068871 | Heterozygous | Tier2 | Not reviewed | 1.3 | — |
| **EXOSC8** | p.Pro87His | chr13:37005941 | rs117135638 | Heterozygous | Tier3 | Benign | 25.5 | 0.662 |
| **EYA2** | c.-11+2233G>A | chr20:46897220 | rs746126704 | Heterozygous | Tier2 | Not reviewed | 18.4 | — |
| **F13A1** | c.1908+5631T>C | chr6:6161827 | rs6928884 | Heterozygous | Tier2 | Not reviewed | 6.9 | — |
| **F5** | p.Leu1397Phe | chr1:169540901 | rs13306334 | Heterozygous | Tier3 | Benign/Likely benign | 0.7 | 0.033 |
| **FAAH2** | p.Arg42Trp | chrX:57286949 | rs146874627 | Heterozygous | Tier2 | Benign | 13.3 | 0.103 |
| **FAAP24** | p.Ser126Phe | chr19:32974193 | rs36017455 | Heterozygous | **Tier1** | Benign | 25.7 | — |
| **FAHD1** | p.Ala102Ser | chr16:1827542 | rs200357596 | Heterozygous | **Tier1** | Not reviewed | 27.2 | 0.715 |
| **FAM135A** | p.Asp619Gly | chr6:70524940 | rs1366288688 | Heterozygous | Tier2 | Not reviewed | 3.7 | 0.073 |
| **FAM50A** | p.Ala137Val | chrX:154446528 | rs149558328 | Heterozygous | Tier2 | Not reviewed | 0.1 | 0.007 |
| **FASN** | p.Arg2125Trp | chr17:82081634 | rs141141382 | Heterozygous | Tier2 | Likely benign | 22.7 | 0.088 |
| **FAT1** | p.Met1149Thr | chr4:186663433 | rs77280820 | Heterozygous | Tier2 | Likely benign | 20.7 | 0.104 |
| **FCGR1A** | p.Asp324Asn | chr1:149791362 | rs1050204 | Heterozygous | Tier2 | Not reviewed | 7.1 | 0.049 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **FCGR3B** | p.Ser36Arg | chr1:161629989 | rs200688856 | Heterozygous | Tier2 | Not reviewed | 0.6 | 0.016 |
| **FHIT** | c.104-75688A>G | chr3:60089840 | rs115894613 | Heterozygous | Tier2 | Not reviewed | 1.7 | — |
| **FHIT** | c.103+176862del | chr3:60359998 | chr3:60359998 | Heterozygous | Tier2 | Not reviewed | — | — |
| **FHIT** | c.103+86860del | chr3:60450000 | chr3:60450000 | Heterozygous | Tier2 | Not reviewed | — | — |
| **FHIT** | c.-17-10312G>C | chr3:60547291 | rs114893898 | Heterozygous | **Tier1** | Not reviewed | 9.7 | — |
| **FLG** | p.Glu2074Gln | chr1:152308666 | rs141120727 | Heterozygous | Tier3 | Benign/Likely benign | 0.0 | 0.033 |
| **FLG** | p.His1880Asn | chr1:152309248 | rs149817134 | Heterozygous | Tier3 | Benign/Likely benign | 20.9 | 0.076 |
| **FLG** | p.Thr581Ser | chr1:152313145 | rs145627745 | Heterozygous | Tier3 | Benign/Likely benign | 0.0 | 0.028 |
| **FLG2** | p.Tyr881Ser | chr1:152355144 | rs79239476 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.016 |
| **FLG2** | p.Ser877Gly | chr1:152355157 | rs144938115 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.012 |
| **FOXO1** | p.Gly104Ala | chr13:40665902 | rs963605898 | Heterozygous | Tier2 | Likely benign | 12.6 | 0.282 |
| **FOXO3** | c.621+9_621+18del | chr6:108569989 | chr6:108569989 | Heterozygous | Tier2 | Not reviewed | — | — |
| **FOXO3** | c.621+10_621+18del | chr6:108569990 | chr6:108569990 | Heterozygous | Tier2 | Not reviewed | — | — |
| **FTCD** | p.Glu128Lys | chr21:46151966 | rs61729373 | Heterozygous | Tier2 | Benign | 20.2 | 0.211 |
| **FZR1** | p.Asn143= | chr19:3527021 | rs138835338 | Heterozygous | Tier2 | Benign | 0.1 | — |
| **GABRA1** | c.*1381T>C | chr5:161898803 | rs187522588 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 13.8 | — |
| **GABRB3** | c.240+5408G>A | chr15:26766994 | rs3212335 | Heterozygous | Tier2 | Not reviewed | 5.9 | — |
| **GAN** | c.1612+12A>G | chr16:81377340 | rs150344737 | Heterozygous | Tier2 | Conflicting classifications of | 0.5 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| | | | | | | pathogenicity | | |
| **GATD3** | p.Val6Ala | chr21:44133714 | rs968714 | Heterozygous | Tier2 | Not reviewed | 0.7 | 0.007 |
| **GATD3** | p.Ser145Arg | chr21:44202056 | rs6518351 | Heterozygous | Tier2 | Not reviewed | 15.6 | — |
| **GDAP1** | c.-18+8599A>G | chr8:74330187 | rs2975212 | Heterozygous | Tier2 | Not reviewed | 0.6 | — |
| **GDAP1** | c.-18+12339A>G | chr8:74333927 | rs2981271 | Heterozygous | Tier2 | Not reviewed | 0.3 | — |
| **GDAP1** | c.-18+14480A>G | chr8:74336068 | rs2975206 | Heterozygous | Tier2 | Not reviewed | 17.1 | — |
| **GFM1** | p.Val664Ile | chr3:158690243 | rs62288347 | Heterozygous | Tier3 | Benign/Likely benign | 22.6 | 0.181 |
| **GJC2** | p.Met147Leu | chr1:228158197 | chr1:228158197 | Heterozygous | Tier2 | Not reviewed | 22.8 | 0.445 |
| **GLRA3** | p.Ala34Glu | chr4:174788914 | rs748400564 | Heterozygous | Tier2 | Uncertain significance | 21.9 | 0.180 |
| **GNAO1** | c.*713del | chr16:56356775 | chr16:56356775 | Heterozygous | Tier2 | Not reviewed | — | — |
| **GNB2** | c.-89-715T>C | chr7:100675462 | rs117960080 | Heterozygous | Tier2 | Not reviewed | 21.5 | — |
| **GNPAT** | c.569-11del | chr1:231265279 | rs199905093 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | — | — |
| **GP5** | p.Met289Ile | chr3:194397416 | rs142440028 | Heterozygous | Tier2 | Not reviewed | 17.3 | 0.027 |
| **GPAT2** | p.Val555Ile | chr2:96024451 | rs775199489 | Homozygous | Tier2 | Uncertain significance | 10.1 | 0.096 |
| **GPER1** | p.Ser373Arg | chr7:1092847 | rs138657567 | Heterozygous | Tier2 | Not reviewed | 0.7 | 0.091 |
| **GPNMB** | c.224-16C>G | chr7:23254153 | rs199346 | Heterozygous | Tier2 | Not reviewed | 0.8 | — |
| **GRIN2A** | c.1122+16340T>C | chr16:9874646 | rs7204266 | Heterozygous | Tier2 | Not reviewed | 4.2 | — |
| **GRIN3B** | p.Pro736Ala | chr19:1007863 | rs144432988 | Heterozygous | Tier2 | Uncertain significance | 24.4 | 0.221 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **GUF1** | p.Arg250Cys | chr4:44686523 | rs116062572 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 22.1 | 0.035 |
| **HACD4** | p.Leu91Phe | chr9:21016010 | rs61736077 | Heterozygous | Tier2 | Not reviewed | 24.9 | 0.219 |
| **HEPACAM** | c.804-242G>C | chr11:124923060 | rs11605766 | Heterozygous | **Tier1** | Likely benign | 14.2 | — |
| **HIVEP2** | c.-640-6984A>G | chr6:142844031 | chr6:142844031 | Heterozygous | Tier2 | Not reviewed | 16.8 | — |
| **HLA-A** | p.Pro217Ala | chr6:29944151 | rs1059563 | Heterozygous | Tier2 | no classification for the single variant | 15.0 | 0.095 |
| **HLA-A** | p.Ile218Val | chr6:29944154 | rs9260179 | Heterozygous | Tier2 | no classification for the single variant | 0.0 | 0.016 |
| **HLA-A** | c.*574G>A | chr6:29948263 | rs9256984 | Homozygous | Tier2 | Uncertain significance | 3.5 | — |
| **HLA-DQA1** | c.-355-2A>G | chr6:32631940 | rs9272089 | Heterozygous | Tier2 | Not reviewed | 9.9 | — |
| **HLA-DQA1** | p.Glu73Asp | chr6:32641446 | rs9272697 | Heterozygous | Tier2 | Not reviewed | 8.3 | 0.031 |
| **HLA-DQA1** | p.Phe74Leu | chr6:32641447 | rs9272698 | Homozygous | Tier2 | Not reviewed | 13.2 | 0.097 |
| **HLA-DQB1** | p.Thr103Asn | chr6:32664869 | rs9274387 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.158 |
| **HLA-DRB5** | p.Val265Leu | chr6:32517747 | rs77867566 | Heterozygous | Tier2 | Not reviewed | 2.0 | 0.218 |
| **HLA-DRB5** | p.Val265Leu | chr6:32517747 | rs77867566 | Heterozygous | Tier2 | Not reviewed | 2.0 | 0.212 |
| **HLA-DRB5** | p.His260Gln | chr6:32518061 | rs112835078 | Homozygous | Tier2 | Not reviewed | 1.0 | 0.307 |
| **HLA-DRB5** | p.Lys127Gln | chr6:32519643 | rs138786370 | Heterozygous | Tier2 | Not reviewed | 0.3 | 0.025 |
| **HLA-DRB5** | c.371-392G>A | chr6:32520043 | rs796105724 | Heterozygous | **Tier1** | Not reviewed | 8.4 | — |
| **HLA-DRB5** | c.371-393T>G | chr6:32520044 | rs796784269 | Heterozygous | **Tier1** | Not reviewed | 2.8 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **HLA-DRB5** | p.His110Tyr | chr6:32521947 | rs142908883 | Heterozygous | Tier2 | Not reviewed | 0.3 | 0.093 |
| **HLA-DRB5** | p.Ala103Gly | chr6:32521967 | rs1059598 | Heterozygous | Tier2 | Not reviewed | 0.1 | 0.109 |
| **HLA-DRB5** | p.Ala102Gly | chr6:32521970 | rs138927633 | Heterozygous | Tier2 | Not reviewed | 15.1 | 0.118 |
| **HLA-DRB5** | p.Asp99Glu | chr6:32521978 | rs41559913 | Homozygous | Tier2 | Not reviewed | 0.0 | 0.033 |
| **HLA-DRB5** | p.Asp99Gly | chr6:32521979 | rs41545413 | Heterozygous | Tier2 | Not reviewed | 2.0 | 0.193 |
| **HLA-DRB5** | p.Asp99His | chr6:32521980 | rs41547217 | Homozygous | Tier2 | Not reviewed | 0.2 | 0.177 |
| **HLA-DRB5** | c.101-1G>A | chr6:32522175 | rs201576424 | Heterozygous | **Tier1** | Not reviewed | 33.0 | — |
| **HLA-DRB5** | c.100+17G>A | chr6:32530108 | rs774541765 | Heterozygous | **Tier1** | Not reviewed | 6.7 | — |
| **HLA-DRB5** | c.-123C>A | chr6:32530347 | rs1188570510 | Heterozygous | Tier2 | Not reviewed | 4.5 | — |
| **HLCS** | c.1438-1318C>T | chr21:36931751 | rs2409866 | Heterozygous | Tier2 | Not reviewed | 1.1 | — |
| **HOXA13** | p.Pro59Ala | chr7:27199903 | rs747262113 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 19.3 | 0.307 |
| **HOXC10** | p.Ser156Cys | chr12:53985726 | rs149717368 | Heterozygous | Tier2 | Uncertain significance | 24.7 | 0.527 |
| **HPSE** | p.Phe236Cys | chr4:83310857 | rs61755719 | Heterozygous | Tier2 | Likely benign | 17.9 | 0.051 |
| **HSD17B1** | p.Ser223Asn | chr17:42554533 | rs146195480 | Heterozygous | Tier2 | Not reviewed | 0.9 | 0.067 |
| **HSD17B6** | p.Ile249Val | chr12:56787133 | rs146253468 | Heterozygous | Tier2 | Uncertain significance | 0.0 | 0.135 |
| **HSPA1B** | p.Arg311Ter | chr6:31828881 | rs1206932739 | Heterozygous | Tier2 | Not reviewed | 36.0 | — |
| **HUWE1** | p.Arg2992His | chrX:53551387 | rs782317854 | Heterozygous | Tier2 | Uncertain significance | 26.4 | 0.255 |
| **ICOSLG** | p.Val128Ile | chr21:44236891 | rs11558819 | Heterozygous | Tier2 | Benign/Likely benign | 3.2 | 0.042 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **IFI27** | p.Val26Ile | chr14:94111758 | rs775606082 | Heterozygous | Tier2 | Not reviewed | 2.5 | 0.015 |
| **IFIT3** | p.Leu442Arg | chr10:89339980 | rs116926108 | Heterozygous | Tier2 | Not reviewed | 23.4 | 0.207 |
| **IFNA10** | p.Val166Ile | chr9:21206602 | rs111996205 | Heterozygous | Tier2 | Not reviewed | 0.3 | 0.058 |
| **IFNA16** | p.Thr124Ile | chr9:21216935 | rs28368161 | Heterozygous | Tier3 | Benign | 0.0 | 0.001 |
| **IFT122** | p.Arg1043= | chr3:129514530 | rs76881473 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 1.1 | — |
| **IGHV4-4** | chr14:106012117 | chr14:106012117 | rs539822603 | Heterozygous | Tier2 | Not reviewed | 33.0 | — |
| **IL17RC** | p.Ala324Thr | chr3:9928397 | rs115461448 | Heterozygous | Tier3 | Benign | 23.2 | 0.091 |
| **IL17RC** | c.1522+1G>C | chr3:9932859 | rs148575246 | Heterozygous | **Tier1** | Benign/Likely benign | 33.0 | — |
| **IL1RAPL1** | c.82+105560T>C | chrX:28894985 | rs28826293 | Heterozygous | Tier2 | Not reviewed | 2.2 | — |
| **IL1RAPL1** | c.82+224347A>G | chrX:29013772 | rs12843979 | Homozygous | Tier2 | Not reviewed | 1.1 | — |
| **INO80D** | p.Arg299Gln | chr2:206056266 | rs144619437 | Heterozygous | Tier2 | Benign | 21.7 | 0.048 |
| **INVS** | c.273+37788A>G | chr9:100164337 | rs863668 | Homozygous | Tier2 | Not reviewed | 2.7 | — |
| **IQCB1** | p.Glu481Lys | chr3:121772683 | rs140630401 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 26.7 | 0.556 |
| **ITGA4** | p.Val824Ala | chr2:181529581 | rs1143675 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 23.4 | 0.104 |
| **ITGA4** | c.*2362C>T | chr2:181537889 | rs148226735 | Heterozygous | Tier2 | Uncertain significance | 7.2 | — |
| **ITGB3BP** | c.254+4779G>A | chr1:63473985 | rs1190436510 | Homozygous | Tier2 | Not reviewed | 7.2 | — |
| **ITGB8** | c.-2181_-2180del | chr7:20329608 | chr7:20329608 | Heterozygous | Tier2 | Not reviewed | — | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **ITPR2** | p.Ser685= | chr12:26657844 | rs2291264 | Heterozygous | Tier3 | Benign | 0.3 | — |
| **KANSL2** | p.Met370Ile | chr12:48660483 | rs780890886 | Heterozygous | Tier2 | Uncertain significance | 22.4 | 0.165 |
| **KCNB1** | p.Ser857Asn | chr20:49372990 | rs34280195 | Heterozygous | Tier3 | Benign | 23.2 | 0.381 |
| **KCNB1** | p.Thr616Ser | chr20:49373713 | rs2229006 | Heterozygous | Tier3 | Benign | 15.7 | 0.110 |
| **KCND3** | c.1107-86635A>G | chr1:111873741 | rs2788612 | Homozygous | Tier2 | Not reviewed | 18.2 | — |
| **KCNK13** | p.Gly331Arg | chr14:90184767 | rs116117726 | Heterozygous | Tier2 | Not reviewed | 19.9 | 0.085 |
| **KCNMB2** | c.-68+29835C>T | chr3:178302841 | rs1357843772 | Heterozygous | Tier2 | Not reviewed | 6.9 | — |
| **KCNU1** | p.Gln1119Arg | chr8:36935826 | rs756494970 | Heterozygous | Tier2 | Uncertain significance | 2.3 | 0.009 |
| **KDM6B** | p.Val209Leu | chr17:7846654 | rs148641957 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 18.4 | 0.029 |
| **KIAA1191** | c.-276A>G | chr5:176361710 | rs1768029962 | Heterozygous | Tier2 | Not reviewed | 20.7 | — |
| **LAMA5** | p.Pro3536Ser | chr20:62310306 | rs41307203 | Heterozygous | Tier3 | Benign | 8.0 | 0.147 |
| **LAMA5** | p.Ala3197= | chr20:62311964 | rs114698761 | Heterozygous | Tier2 | Benign | 0.2 | — |
| **LAMA5** | p.Arg3165Gln | chr20:62312183 | rs140181393 | Heterozygous | Tier3 | Benign | 12.4 | 0.113 |
| **LAMA5** | p.Arg2123His | chr20:62322147 | rs374836875 | Heterozygous | Tier2 | Benign/Likely benign | 16.6 | 0.026 |
| **LAMA5** | p.Arg1903Lys | chr20:62324140 | rs116895933 | Heterozygous | Tier3 | Benign/Likely benign | 1.0 | 0.054 |
| **LAMA5** | p.Pro1260Ala | chr20:62330817 | rs780286336 | Heterozygous | Tier2 | Uncertain significance | 15.3 | 0.108 |
| **LAMB2** | p.Ala1765Thr | chr3:49121330 | rs74951356 | Heterozygous | Tier3 | Benign/Likely benign | 13.9 | 0.046 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **LAMB2** | p.Arg1436His | chr3:49122970 | rs377526198 | Heterozygous | Tier2 | Uncertain significance | 21.3 | 0.053 |
| **LAMB2** | p.Leu769= | chr3:49126004 | rs147076626 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 5.5 | — |
| **LCA5** | c.-398G>C | chr6:79537371 | rs370115829 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 7.2 | — |
| **LDAH** | p.Arg151Gln | chr2:20774826 | rs114010802 | Heterozygous | Tier2 | Benign | 0.9 | 0.047 |
| **LDHAL6A** | c.592+1G>A | chr11:18475640 | rs145833201 | Heterozygous | **Tier1** | Not reviewed | 31.0 | — |
| **LEP** | c.*653A>C | chr7:128255416 | rs201341056 | Heterozygous | Tier2 | Uncertain significance | 0.8 | — |
| **LIFR** | p.Arg780Cys | chr5:38485978 | rs553912841 | Heterozygous | Tier2 | Uncertain significance | 28.6 | 0.166 |
| **LIFR** | p.Tyr762His | chr5:38489129 | rs200270315 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 24.4 | 0.240 |
| **LIFR** | p.Ser723fs | chr5:38489995 | chr5:38489995 | Heterozygous | Tier2 | Not reviewed | — | — |
| **LIFR** | p.Ser723fs | chr5:38489999 | chr5:38489999 | Heterozygous | Tier2 | Not reviewed | — | — |
| **LILRA6** | p.Gln350Ter | chr19:54240484 | rs2361802 | Heterozygous | Tier2 | Not reviewed | 22.2 | — |
| **LILRA6** | p.Arg205Trp | chr19:54241621 | rs1052973 | Heterozygous | Tier2 | Not reviewed | 16.3 | 0.015 |
| **LILRA6** | p.Ser122Asn | chr19:54241869 | rs1132595 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRA6** | p.Ala120Val | chr19:54241875 | rs1132593 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRA6** | p.Ala120Ser | chr19:54241876 | rs1132591 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRA6** | p.Met117Thr | chr19:54242031 | rs62133127 | Heterozygous | Tier2 | Not reviewed | 12.4 | 0.099 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **LILRA6** | p.His20Arg | chr19:54242530 | rs201396172 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRB3** | p.Tyr350Ter | chr19:54220736 | rs16985497 | Homozygous | Tier2 | Not reviewed | 23.2 | — |
| **LILRB3** | p.Tyr350His | chr19:54220738 | rs1052987 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRB3** | p.Gly349Trp | chr19:54220741 | rs1052985 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRB3** | p.Arg348Trp | chr19:54220744 | rs140963490 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRB3** | p.Thr318Ala | chr19:54221086 | rs563416061 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **LILRB3** | p.Ile317Met | chr19:54221087 | rs529098038 | Homozygous | Tier2 | Not reviewed | 8.8 | — |
| **LITAF** | c.-5-858C>A | chr16:11557593 | rs9923255 | Heterozygous | Tier2 | Not reviewed | 1.5 | — |
| **LNPK** | c.358-1181T>C | chr2:175965770 | chr2:175965770 | Heterozygous | Tier2 | Not reviewed | 8.1 | — |
| **LPA** | p.Tyr2023Cys | chr6:160531784 | rs41267807 | Heterozygous | Tier3 | Not reviewed | 25.9 | 0.614 |
| **LRRK2** | p.Pro1542Ser | chr12:40314059 | rs33958906 | Heterozygous | Tier3 | Benign/Likely benign | 24.7 | 0.228 |
| **LTF** | c.1357+398C>T | chr3:46446042 | rs766222508 | Heterozygous | Tier2 | Not reviewed | 5.0 | — |
| **MACF1** | c.21959-173C>T | chr1:39479625 | rs2275767 | Heterozygous | Tier2 | Benign | 3.4 | — |
| **MACROD1** | c.665-35G>C | chr11:63999798 | rs202065074 | Heterozygous | Tier2 | Not reviewed | 5.8 | — |
| **MAGI2** | c.1046-308C>T | chr7:78369521 | rs7783086 | Heterozygous | Tier2 | Benign | 0.3 | — |
| **MAK** | c.-254G>C | chr6:10838527 | rs72821595 | Heterozygous | Tier2 | Uncertain significance | 6.3 | — |
| **MAPT** | p.Gln305Arg | chr17:45983493 | rs63750072 | Heterozygous | Tier3 | Benign/Likely benign | 22.3 | 0.079 |
| **MC1R** | p.Asp294His | chr16:89920138 | rs1805009 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity\|risk factor | 26.9 | 0.608 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **MCPH1** | c.-1527_-1526del | chr8:6405127 | chr8:6405127 | Heterozygous | Tier2 | Not reviewed | — | — |
| **MDM4** | p.Lys374Gln | chr1:204549329 | rs41299595 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 21.4 | 0.047 |
| **MEF2C** | c.638-1909G>A | chr5:88733810 | rs412458 | Heterozygous | Tier2 | Not reviewed | 0.3 | — |
| **MEF2C** | c.-528_-526del | chr5:88883338 | chr5:88883338 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | — | — |
| **METTL9** | c.-251C>G | chr16:21599483 | rs1478461 | Homozygous | Tier2 | Not reviewed | 7.0 | — |
| **MFSD12** | p.Arg472Gln | chr19:3544814 | rs200371465 | Heterozygous | Tier2 | Not reviewed | 22.4 | 0.101 |
| **MGST3** | p.Trp139Ser | chr1:165655461 | rs199915214 | Heterozygous | Tier2 | Not reviewed | 29.2 | 0.496 |
| **MID1IP1** | p.Gly182Asp | chrX:38805491 | rs143008624 | Heterozygous | **Tier1** | Benign | 26.1 | 0.630 |
| **MMADHC** | p.Lys29Asn | chr2:149582194 | rs61750442 | Heterozygous | Tier2 | Benign/Likely benign | 23.3 | 0.327 |
| **MMP19** | p.Arg456Gln | chr12:55837196 | rs150724096 | Heterozygous | Tier2 | Likely benign | 21.5 | 0.097 |
| **MPDZ** | p.Arg1229Gln | chr9:13147603 | rs61753783 | Heterozygous | Tier2 | Uncertain significance | 28.6 | 0.384 |
| **MPL** | c.1654-10T>A | chr1:43352508 | rs200460456 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 10.9 | — |
| **MPV17** | c.*413A>C | chr2:27309499 | rs575276701 | Heterozygous | Tier2 | Uncertain significance | 4.5 | — |
| **MRPL21** | c.396+31G>A | chr11:68896484 | rs76789024 | Heterozygous | Tier2 | Not reviewed | 6.4 | — |
| **MRPL23** | p.Ala122Thr | chr11:1956322 | rs12812 | Heterozygous | Tier2 | Not reviewed | 0.0 | 0.008 |
| **MRPL23** | p.Arg162Ser | chr11:1972745 | rs190063 | Heterozygous | Tier2 | Not reviewed | 8.7 | 0.011 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **MRPL34** | p.Ala74Val | chr19:17306321 | rs201529220 | Heterozygous | Tier2 | Not reviewed | 20.7 | 0.033 |
| **MSLN** | p.Arg557His | chr16:768452 | rs150425699 | Heterozygous | Tier3 | Benign | 16.4 | 0.034 |
| **MSRA** | c.211+6549G>A | chr8:10214450 | rs955247088 | Heterozygous | Tier2 | Not reviewed | 16.8 | — |
| **MTHFD1** | p.Lys134Arg | chr14:64415662 | rs1950902 | Heterozygous | Tier3 | Benign | 20.0 | 0.237 |
| **MTMR3** | c.*63G>T | chr22:30025864 | rs746220322 | Heterozygous | Tier2 | Not reviewed | 16.1 | — |
| **MTR** | p.Asp314Asn | chr1:236826841 | rs2229274 | Heterozygous | Tier3 | Benign | 22.5 | 0.120 |
| **MTR** | c.*4347A>T | chr1:236901991 | rs767109806 | Heterozygous | Tier2 | Uncertain significance | 1.2 | — |
| **MUC16** | p.Arg12256Gln | chr19:8916887 | rs139868374 | Heterozygous | Tier2 | Benign | 14.7 | 0.059 |
| **MX1** | p.Arg265Gln | chr21:41441779 | rs145851733 | Heterozygous | Tier2 | Uncertain significance | 14.0 | 0.123 |
| **MXRA8** | p.Asp396Asn | chr1:1354066 | rs150058708 | Heterozygous | Tier2 | Likely benign | 23.3 | 0.101 |
| **MYO6** | c.*1324A>G | chr6:75916336 | rs191987417 | Heterozygous | Tier2 | Uncertain significance | 10.5 | — |
| **MYOF** | c.4719+179G>A | chr10:93333579 | rs12782443 | Heterozygous | Tier2 | Not reviewed | 12.6 | — |
| **MYOM1** | c.1174+2180C>T | chr18:3171758 | rs12965146 | Heterozygous | Tier2 | Not reviewed | 3.4 | — |
| **NAIP** | p.Val535Met | chr5:70985446 | rs1423904967 | Homozygous | Tier2 | Not reviewed | 5.1 | 0.035 |
| **NFIA** | c.96+3376C>G | chr1:60868842 | rs1375339493 | Heterozygous | Tier2 | Not reviewed | 15.7 | — |
| **NFIA** | c.626-26539G>A | chr1:61305973 | rs2806434 | Heterozygous | Tier2 | Not reviewed | 1.8 | — |
| **NIPSNAP2** | c.712+1G>T | chr7:55994989 | rs748801121 | Heterozygous | **Tier1** | Not reviewed | 33.0 | — |
| **NLRC4** | p.Ala883Thr | chr2:32235536 | rs780229500 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 3.4 | 0.018 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **NLRC4** | p.Tyr594Asp | chr2:32250084 | chr2:32250084 | Heterozygous | Tier2 | Not reviewed | 22.4 | 0.261 |
| **NLRP2** | p.Thr516Ala | chr19:54983244 | rs61735082 | Heterozygous | Tier3 | Likely benign | 0.0 | 0.070 |
| **NLRP2** | p.Glu522Gly | chr19:54983263 | rs61735083 | Heterozygous | Tier3 | Likely benign | 8.7 | 0.097 |
| **NLRP4** | p.Gly638Arg | chr19:55861441 | rs143589259 | Heterozygous | Tier2 | Not reviewed | 1.9 | 0.201 |
| **NLRP9** | p.Arg899Cys | chr19:55711948 | rs148511077 | Heterozygous | Tier3 | Benign | 1.7 | 0.128 |
| **NME2** | c.-5+1G>A | chr17:51166498 | rs115848216 | Heterozygous | Tier2 | Not reviewed | 16.9 | — |
| **NPC2** | c.441+1G>A | chr14:74480701 | rs140130028 | Heterozygous | **Tier1** | Conflicting classifications of pathogenicity | 23.2 | 0.207 |
| **NPHS1** | c.-57G>T | chr19:35851894 | rs140404523 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 5.7 | — |
| **NPY4R** | p.Cys240Arg | chr10:46461918 | rs1352903918 | Heterozygous | Tier3 | Benign | 16.7 | — |
| **NQO1** | p.Pro187Ser | chr16:69711242 | rs1800566 | Heterozygous | Tier2 | Uncertain significance | 24.2 | 0.366 |
| **NR4A1** | p.Thr143Met | chr12:52054756 | rs373652442 | Heterozygous | Tier2 | Uncertain significance | 24.8 | 0.528 |
| **NRP1** | p.Asp636Asn | chr10:33197668 | rs145594886 | Heterozygous | Tier2 | Uncertain significance | 22.9 | 0.392 |
| **NRXN1** | c.4128+39765G>A | chr2:50013506 | rs3796036 | Homozygous | Tier2 | Not reviewed | 0.1 | — |
| **NRXN1** | c.3365-51910A>G | chr2:50288880 | rs1356888 | Heterozygous | Tier2 | Not reviewed | 1.8 | — |
| **NRXN1** | c.-1273G>A | chr2:51032332 | rs186103615 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 14.1 | — |
| **NRXN3** | c.2275+65009A>T | chr14:78875353 | rs866714403 | Heterozygous | Tier2 | Not reviewed | 15.4 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **NTNG1** | c.1391-11644G>A | chr1:107468967 | rs11185120 | Heterozygous | Tier2 | Not reviewed | 0.8 | — |
| **OGFOD3** | c.380+1G>A | chr17:82411454 | rs150938774 | Heterozygous | **Tier1** | Not reviewed | 34.0 | — |
| **ONECUT1** | c.1105+3262C>T | chr15:52785518 | rs34939760 | Heterozygous | Tier2 | Not reviewed | 1.5 | — |
| **OPA3** | c.*1070C>T | chr19:45552444 | rs146989248 | Heterozygous | Tier2 | Uncertain significance | 1.6 | — |
| **ORAI1** | p.Ser218Gly | chr12:121641389 | rs3741596 | Heterozygous | Tier3 | Benign | 1.1 | — |
| **ORC1** | p.Ala372Val | chr1:52389289 | rs3087476 | Heterozygous | Tier3 | Benign | 12.2 | 0.122 |
| **P3H2** | c.824-1035T>A | chr3:189990073 | rs55848820 | Homozygous | Tier2 | Not reviewed | 9.0 | — |
| **P3H3** | p.Arg199Ter | chr12:6829955 | rs1225283465 | Heterozygous | Tier2 | Not reviewed | 35.0 | — |
| **P4HA2** | c.-19+2739_-19+2740dup | chr5:132225051 | chr5:132225051 | Heterozygous | Tier2 | Not reviewed | — | — |
| **PACS2** | p.Pro725Leu | chr14:105391685 | rs147461490 | Heterozygous | Tier3 | Benign | 23.6 | 0.157 |
| **PAK4** | p.Arg153Gln | chr19:39173171 | rs754471124 | Heterozygous | Tier2 | Uncertain significance | 22.3 | 0.173 |
| **PALLD** | c.1965-12907G>A | chr4:168878015 | rs780516159 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 10.2 | 0.062 |
| **PAPSS1** | c.1736+4598T>A | chr4:107627033 | chr4:107627033 | Heterozygous | Tier2 | Not reviewed | 18.6 | — |
| **PARPBP** | p.Asn81Asp | chr12:102148317 | chr12:102148317 | Heterozygous | Tier2 | Not reviewed | 15.6 | 0.016 |
| **PCDH15** | c.4368-1931C>T | chr10:53822161 | rs111033445 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 0.5 | — |
| **PCDHA9** | p.Gln534Glu | chr5:140850095 | rs139719626 | Heterozygous | Tier2 | Not reviewed | 15.4 | 0.119 |
| **PCDHGA5** | p.Ser184Arg | chr5:141364882 | rs199576947 | Heterozygous | Tier2 | Not reviewed | 16.4 | 0.064 |
| **PGK1** | p.Lys133Glu | chrX:78114140 | chrX:78114140 | Heterozygous | **Tier1** | Not reviewed | 22.7 | 0.543 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **PHIP** | c.3379+491A>G | chr6:78965212 | rs10943608 | Heterozygous | Tier2 | Not reviewed | 1.3 | — |
| **PHKB** | p.Met185Ile | chr16:47515562 | rs56257827 | Heterozygous | Tier3 | Benign/Likely benign | 26.2 | 0.773 |
| **PI4KA** | p.Ile339Val | chr22:20811023 | rs61752248 | Heterozygous | Tier3 | Likely benign | 23.6 | 0.016 |
| **PI4KA** | c.530-124G>T | chr22:20820024 | rs165691 | Heterozygous | Tier3 | Benign | 0.4 | — |
| **PIEZO1** | c.65-6410A>G | chr16:88755889 | rs3743975 | Homozygous | Tier2 | Not reviewed | 1.8 | — |
| **PIKFYVE** | c.-10+5G>A | chr2:208266420 | rs541362896 | Heterozygous | Tier3 | Likely benign | 18.1 | — |
| **PKD2** | c.*1010G>A | chr4:88076704 | rs144024108 | Heterozygous | Tier2 | Uncertain significance | 0.1 | — |
| **PKHD1** | c.6122-5904A>G | chr6:51918480 | rs12199884 | Heterozygous | Tier2 | Not reviewed | 1.3 | — |
| **PKP4** | c.-5-16141T>C | chr2:158517039 | rs1436595521 | Heterozygous | Tier2 | Not reviewed | 20.1 | — |
| **PLA2G4E** | p.Val716Met | chr15:41984589 | rs774426285 | Heterozygous | Tier2 | Uncertain significance | 1.7 | 0.017 |
| **PLA2G4F** | c.601+417C>T | chr15:42152271 | rs116885045 | Heterozygous | Tier2 | Not reviewed | 0.0 | — |
| **PLB1** | c.-114+4401G>A | chr2:28461598 | rs56116810 | Heterozygous | **Tier1** | Not reviewed | 3.5 | — |
| **PLCD1** | p.Asp231Asn | chr3:38011313 | rs61755441 | Heterozygous | Tier3 | Benign | 19.9 | 0.181 |
| **PLEKHG2** | p.Pro453Arg | chr19:39420811 | rs34975345 | Homozygous | Tier3 | Benign | 22.5 | 0.083 |
| **PLEKHG2** | p.Ala770Thr | chr19:39423362 | rs111487768 | Heterozygous | Tier2 | Benign/Likely benign | 25.0 | 0.378 |
| **PLK4** | p.Gly6Glu | chr4:127881151 | rs149003893 | Heterozygous | **Tier1** | Conflicting classifications of pathogenicity | 28.5 | 0.356 |
| **PMS2** | p.Ala520Val | chr7:5987206 | rs63751300 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 1.4 | 0.090 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **PNKP** | c.1127-8C>T | chr19:49862113 | rs3739203 | Heterozygous | Tier3 | Benign/Likely benign | 5.3 | — |
| **PNKP** | p.Pro20Ser | chr19:49867147 | rs3739168 | Heterozygous | Tier2 | Benign/Likely benign | 17.9 | 0.035 |
| **POLDIP2** | p.Ala36Val | chr17:28357342 | rs11555978 | Heterozygous | Tier3 | Likely benign | 15.0 | — |
| **PPM1D** | p.Ser376Phe | chr17:60656708 | rs61739281 | Heterozygous | Tier2 | Uncertain significance | 24.0 | 0.194 |
| **PRICKLE1** | c.-48-11679G>C | chr12:42484243 | rs538816869 | Heterozygous | Tier2 | Not reviewed | 9.5 | — |
| **PRKD1** | c.33+60419_33+60420del | chr14:29987294 | chr14:29987294 | Heterozygous | Tier2 | Not reviewed | — | — |
| **PRKD1** | c.33+31108A>G | chr14:30016602 | rs1360253541 | Heterozygous | Tier2 | Not reviewed | 19.2 | — |
| **PRKDC** | c.-766G>A | chr8:47960892 | rs17334305 | Heterozygous | Tier2 | Uncertain significance | 2.4 | — |
| **PRKN** | c.8-14826C>T | chr6:162458299 | rs771567497 | Heterozygous | Tier2 | Not reviewed | 2.3 | — |
| **PRODH** | c.1251+419G>A | chr22:18919032 | rs449138 | Homozygous | Tier2 | Not reviewed | 0.5 | — |
| **PRODH** | p.Trp185Arg | chr22:18925165 | rs4819756 | Heterozygous | Tier2 | Benign | 7.6 | 0.124 |
| **PROS1** | c.*686G>A | chr3:93873559 | rs550909963 | Heterozygous | Tier2 | Uncertain significance | 0.9 | — |
| **PRR12** | p.Ala649Val | chr19:49596281 | rs371012654 | Heterozygous | Tier2 | Likely benign | 22.3 | 0.202 |
| **PTPRC** | p.Asp123His | chr1:198699632 | rs41269905 | Heterozygous | Tier3 | Benign/Likely benign | 22.8 | 0.070 |
| **PTPRC** | p.Ile296Leu | chr1:198706934 | rs2230606 | Heterozygous | Tier3 | Benign/Likely benign | 0.0 | 0.043 |
| **PTPRC** | p.His570Gln | chr1:198722466 | rs12136658 | Heterozygous | Tier3 | Benign/Likely benign | 0.0 | 0.055 |
| **PUS3** | p.Pro179Ser | chr11:125895633 | rs35681508 | Heterozygous | Tier3 | Benign | 24.5 | 0.262 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **PXDNL** | p.Cys1258Ter | chr8:51372000 | rs117752382 | Heterozygous | Tier2 | Benign | 36.0 | — |
| **PYROXD2** | c.127+23G>C | chr10:98414986 | rs1475191215 | Heterozygous | Tier2 | Not reviewed | 9.4 | — |
| **RAD1** | p.Thr104Ser | chr5:34911809 | rs1805328 | Heterozygous | Tier3 | Benign | 14.9 | 0.055 |
| **RAD51B** | c.756+102812del | chr14:67989999 | chr14:67989999 | Heterozygous | Tier2 | Not reviewed | — | — |
| **RANBP2** | p.Gly2692Asp | chr2:108772543 | rs111958052 | Heterozygous | Tier2 | Likely benign | 13.6 | 0.065 |
| **RAP1GDS1** | p.Val105Met | chr4:98352553 | rs61758810 | Heterozygous | Tier2 | Likely benign | 20.3 | 0.320 |
| **RCC1L** | p.Arg15Trp | chr7:75073695 | rs201728682 | Heterozygous | Tier3 | Benign | 22.9 | — |
| **REN** | c.-211+1G>A | chr1:204190231 | rs76834242 | Heterozygous | Tier2 | Benign | 17.4 | — |
| **RFXAP** | p.Met137Thr | chr13:36819767 | rs193240312 | Heterozygous | Tier3 | Benign/Likely benign | 14.2 | 0.027 |
| **RHCE** | p.Pro103Ser | chr1:25408711 | rs676785 | Homozygous | Tier2 | not provided | 0.0 | 0.015 |
| **RHCE** | p.Asn68Ser | chr1:25408815 | rs1053344 | Homozygous | Tier2 | Not reviewed | 3.7 | 0.043 |
| **RHCE** | p.Leu60Ile | chr1:25408840 | rs181860403 | Homozygous | Tier2 | not provided | 12.0 | 0.057 |
| **RNF152** | p.Lys108Asn | chr18:61816140 | rs369621346 | Heterozygous | Tier2 | Uncertain significance | 18.8 | 0.261 |
| **ROBO1** | c.500-57835_500-57833del | chr3:78804734 | chr3:78804734 | Heterozygous | Tier2 | Not reviewed | — | — |
| **RTN3** | p.Asp501His | chr11:63720003 | rs7936660 | Heterozygous | Tier2 | Not reviewed | 25.6 | 0.236 |
| **RUNX1** | c.59-63960A>G | chr21:34956923 | rs148127564 | Heterozygous | Tier2 | Not reviewed | 10.7 | — |
| **SAA1** | c.91+760T>C | chr11:18267738 | rs111755795 | Heterozygous | Tier2 | Not reviewed | 3.7 | — |
| **SACS** | p.Ile2575= | chr13:23336151 | rs200735789 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 9.0 | — |
| **SCAF4** | p.Phe341Leu | chr21:31696160 | rs117061499 | Heterozygous | Tier2 | Likely benign | 24.1 | 0.096 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **SCN3A** | c.-248+3133G>T | chr2:165200690 | rs116793108 | Heterozygous | Tier2 | Not reviewed | 20.0 | — |
| **SCNN1A** | p.Thr42= | chr12:6374658 | rs138995556 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 2.9 | — |
| **SEC23B** | p.Val426Ile | chr20:18532706 | rs41309927 | Heterozygous | Tier3 | Benign/Likely benign | 18.2 | 0.122 |
| **SEPTIN9** | c.76+23238A>C | chr17:77330435 | rs892964 | Heterozygous | Tier2 | Not reviewed | 2.3 | — |
| **SERPINA1** | c.*80C>T | chr14:94378369 | rs1243163 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 0.1 | — |
| **SERPINC1** | p.Gln337= | chr1:173909694 | rs5878 | Heterozygous | Tier2 | Benign | 9.2 | — |
| **SERPINH1** | c.*228A>G | chr11:75572311 | rs189882479 | Heterozygous | Tier2 | Uncertain significance | 3.5 | — |
| **SETD1B** | p.Ala1488Thr | chr12:121823041 | rs144871159 | Heterozygous | Tier3 | Benign | 10.6 | 0.026 |
| **SETD5** | c.-470_-468del | chr3:9397655 | chr3:9397655 | Homozygous | Tier2 | Not reviewed | — | — |
| **SFXN5** | c.625+7137C>T | chr2:72981121 | rs1363816061 | Heterozygous | Tier2 | Not reviewed | 4.2 | — |
| **SIL1** | c.-10-12945G>A | chr5:139140798 | rs1187156016 | Heterozygous | Tier2 | Not reviewed | 17.6 | — |
| **SIM2** | p.Pro649Ala | chr21:36748033 | rs1469534783 | Heterozygous | Tier2 | Not reviewed | 2.1 | 0.075 |
| **SIMC1** | c.130-22537G>A | chr5:176267117 | rs2560173 | Homozygous | Tier2 | Not reviewed | 3.2 | — |
| **SIRPA** | p.Asp95Glu | chr20:1915304 | rs138283486 | Homozygous | Tier2 | Not reviewed | 0.8 | 0.299 |
| **SIRPA** | p.Leu96Phe | chr20:1915305 | rs149634649 | Homozygous | Tier2 | Not reviewed | 0.0 | 0.072 |
| **SIRPA** | p.Leu96Pro | chr20:1915306 | rs146163282 | Homozygous | Tier2 | Not reviewed | 0.2 | 0.033 |
| **SIRT2** | p.Arg153His | chr19:38889130 | rs144373891 | Heterozygous | Tier2 | Benign | 26.0 | 0.231 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **SLC12A5** | p.Arg86Gly | chr20:46022977 | rs58366483 | Homozygous | Tier2 | Benign/Likely benign | 14.4 | — |
| **SLC12A6** | c.-73+250A>G | chr15:34337082 | rs141533243 | Heterozygous | Tier3 | Likely benign | 15.8 | — |
| **SLC25A15** | c.*1721C>T | chr13:40811388 | rs536553279 | Heterozygous | Tier2 | Uncertain significance | 6.6 | — |
| **SLC27A3** | p.Leu173Pro | chr1:153776015 | rs756978807 | Heterozygous | **Tier1** | Not reviewed | 27.2 | 0.411 |
| **SLC30A7** | c.*3807T>C | chr1:100978664 | chr1:100978664 | Heterozygous | Tier2 | Not reviewed | 15.9 | — |
| **SLC4A2** | p.Arg304Trp | chr7:151066937 | rs371485247 | Heterozygous | Tier2 | Uncertain significance | 26.5 | 0.172 |
| **SLC5A4** | c.886-5G>A | chr22:32233039 | rs9621415 | Heterozygous | **Tier1** | Not reviewed | 14.0 | — |
| **SLC6A4** | c.1818+1348A>G | chr17:30201824 | rs11080121 | Heterozygous | Tier2 | Not reviewed | 5.6 | — |
| **SLC8A1** | c.1808+48_1808+49del | chr2:40428426 | chr2:40428426 | Heterozygous | Tier2 | Not reviewed | — | — |
| **SLCO3A1** | p.Ile560Val | chr15:92147149 | rs141187463 | Heterozygous | Tier2 | Likely benign | 23.0 | 0.180 |
| **SLF2** | c.2513-423A>G | chr10:100938172 | rs531379316 | Heterozygous | Tier2 | Uncertain significance | 0.3 | — |
| **SLIT2** | p.Gly909Asp | chr4:20567262 | rs1020555095 | Heterozygous | **Tier1** | Not reviewed | 32.0 | 0.743 |
| **SMG1** | p.Ala1209Ser | chr16:18863720 | rs137863609 | Heterozygous | Tier3 | Benign | 16.1 | 0.057 |
| **SMPD4** | p.Pro217Leu | chr2:130172358 | rs144353633 | Heterozygous | Tier2 | Not reviewed | 27.6 | 0.303 |
| **SOX5** | c.271-14760del | chr12:23860954 | chr12:23860954 | Heterozygous | Tier2 | Not reviewed | — | — |
| **SP110** | p.Glu538Lys | chr2:230172938 | rs149682257 | Heterozygous | Tier2 | Uncertain significance | 13.9 | 0.267 |
| **SP110** | c.1349-1241T>C | chr2:230179496 | rs967006 | Homozygous | Tier2 | Not reviewed | 3.4 | — |
| **SPAG16** | c.1720+38410G>A | chr2:214187676 | chr2:214187676 | Heterozygous | Tier2 | Not reviewed | 0.7 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **SPG11** | p.Pro2345Leu | chr15:44564664 | rs748206574 | Heterozygous | Tier2 | Uncertain significance | 32.0 | 0.498 |
| **SPIDR** | p.Met316Lys | chr8:47440392 | rs201125764 | Heterozygous | Tier2 | Likely benign | 19.8 | 0.131 |
| **SPRTN** | p.Thr439Ser | chr1:231353206 | rs62617126 | Heterozygous | Tier2 | Benign | 0.1 | 0.003 |
| **SRD5A2** | p.Ala49Thr | chr2:31580756 | rs9282858 | Heterozygous | Tier3 | Benign | 20.9 | — |
| **SSTR4** | p.Ala245Asp | chr20:23036217 | rs35136964 | Heterozygous | Tier2 | Not reviewed | 24.0 | 0.419 |
| **ST3GAL3** | p.Trp70fs | chr1:43829996 | rs71579312 | Homozygous | Tier2 | Not reviewed | — | — |
| **ST3GAL5** | c.-91G>A | chr2:85888996 | rs879798501 | Heterozygous | Tier2 | Uncertain significance | 12.1 | — |
| **STAP1** | c.730-41G>C | chr4:67593219 | rs3816821 | Heterozygous | Tier3 | Benign | 0.3 | — |
| **STOX1** | p.Asn825Ile | chr10:68886270 | rs41278532 | Heterozygous | Tier2 | Not reviewed | 15.5 | 0.214 |
| **SUCLA2** | c.*214+8995G>A | chr13:47896334 | rs7334463 | Homozygous | Tier2 | Not reviewed | 0.0 | — |
| **SUCLA2** | c.*27-16018C>T | chr13:47923153 | rs9591123 | Homozygous | Tier2 | Not reviewed | 1.2 | — |
| **SUMF1** | p.Asn352Lys | chr3:4362213 | rs376126385 | Heterozygous | **Tier1** | Uncertain significance | 23.9 | 0.641 |
| **SYNPR** | c.84+40453T>G | chr3:63319195 | rs73123134 | Heterozygous | **Tier1** | Not reviewed | 6.8 | — |
| **SYT1** | c.*833_*834dup | chr12:79449926 | chr12:79449926 | Heterozygous | Tier2 | Not reviewed | — | — |
| **SYT9** | c.146-23864A>G | chr11:7279175 | chr11:7279175 | Heterozygous | Tier2 | Not reviewed | 16.5 | — |
| **TAC4** | c.106-338G>A | chr17:49844495 | chr17:49844495 | Heterozygous | Tier2 | Not reviewed | 1.7 | — |
| **TANC2** | c.-24+5272A>C | chr17:62972021 | rs763655159 | Heterozygous | Tier2 | Not reviewed | 14.2 | — |
| **TAP1** | p.Met739Val | chr6:32845611 | rs780766118 | Heterozygous | Tier2 | Uncertain significance | 22.8 | 0.440 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **TARS2** | p.His345Arg | chr1:150497543 | rs115390773 | Heterozygous | Tier2 | Benign/Likely benign | 1.6 | 0.067 |
| **TAT** | c.*409G>A | chr16:71567735 | rs572677025 | Heterozygous | Tier2 | Uncertain significance | 0.8 | — |
| **TBP** | p.Gln95del | chr6:170561959 | chr6:170561959 | Heterozygous | Tier3 | Benign | — | — |
| **TCF20** | p.Arg1438Cys | chr22:42210994 | rs771336001 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 24.4 | 0.166 |
| **TCF7L2** | p.Pro500Thr | chr10:113165610 | rs77961654 | Heterozygous | Tier3 | Benign | 22.6 | 0.378 |
| **TECPR2** | p.Lys470dup | chr14:102432106 | chr14:102432106 | Heterozygous | Tier3 | Likely benign | — | — |
| **TECPR2** | p.Ser1092Leu | chr14:102449828 | rs72700618 | Heterozygous | Tier2 | Benign/Likely benign | 27.6 | 0.407 |
| **TEK** | p.Asn438Ile | chr9:27185615 | rs189543659 | Heterozygous | **Tier1** | Not reviewed | 24.3 | 0.208 |
| **TEK** | p.Val600Leu | chr9:27197488 | rs35030851 | Heterozygous | Tier3 | Benign | 17.9 | 0.058 |
| **THSD1** | p.Lys768Arg | chr13:52377667 | rs9536041 | Heterozygous | Tier3 | Benign | 14.7 | 0.092 |
| **TICAM1** | p.Pro367del | chr19:4817277 | chr19:4817277 | Heterozygous | Tier3 | Likely benign | — | — |
| **TIRAP** | p.Ser180Leu | chr11:126292948 | rs8177374 | Heterozygous | Tier3 | Benign | 13.5 | 0.092 |
| **TLK2** | c.1550+842G>A | chr17:62597516 | rs2082566857 | Heterozygous | Tier2 | Not reviewed | 11.1 | — |
| **TLN1** | p.Gln1524Lys | chr9:35707793 | chr9:35707793 | Heterozygous | Tier2 | Not reviewed | 24.8 | 0.325 |
| **TLR1** | p.His305Leu | chr4:38797918 | rs3923647 | Heterozygous | Tier3 | Benign | 17.6 | 0.079 |
| **TLR2** | p.Phe217Ser | chr4:153703557 | rs139227237 | Heterozygous | Tier2 | Benign/Likely benign | 20.2 | 0.526 |
| **TMEM79** | p.Arg127Cys | chr1:156285605 | rs138888317 | Heterozygous | Tier2 | Not reviewed | 25.5 | 0.235 |
| **TNC** | p.Arg1698Gln | chr9:115046442 | rs61734387 | Heterozygous | Tier3 | Benign | 25.6 | 0.185 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **TNC** | p.Arg1066Cys | chr9:115073621 | rs139280264 | Heterozygous | Tier2 | Likely benign | 23.0 | 0.197 |
| **TNR** | c.499+596_499+597del | chr1:175405623 | chr1:175405623 | Heterozygous | Tier2 | Not reviewed | — | — |
| **TOM1** | c.1284+86A>G | chr22:35345870 | rs133399 | Heterozygous | Tier3 | Benign | 2.7 | — |
| **TP53BP1** | p.Glu742Asp | chr15:43456382 | rs150423877 | Heterozygous | Tier2 | Not reviewed | 21.9 | 0.080 |
| **TP53RK** | p.Arg123Gln | chr20:46687147 | rs34983477 | Heterozygous | Tier3 | Benign | 22.8 | 0.126 |
| **TPK1** | c.614-47135A>C | chr7:144500798 | rs1308701958 | Heterozygous | Tier2 | Not reviewed | 12.9 | — |
| **TPM4** | c.133-1860T>G | chr19:16080053 | rs16981095 | Heterozygous | Tier2 | Not reviewed | 3.0 | — |
| **TPTE2** | p.Ser434Trp | chr13:19430469 | rs149205749 | Heterozygous | Tier2 | Not reviewed | 0.5 | 0.199 |
| **TRAV1-2** | chr14:21641263 | chr14:21641263 | chr14:21641263 | Heterozygous | Tier2 | Not reviewed | — | — |
| **TRIO** | c.2046+9360A>G | chr5:14346087 | rs62344980 | Heterozygous | Tier2 | Not reviewed | 4.3 | — |
| **TRPC7** | c.-438_-433dup | chr5:136365689 | chr5:136365689 | Heterozygous | Tier2 | Not reviewed | — | — |
| **TRPM1** | p.Glu1303Lys | chr15:31002793 | rs117855013 | Heterozygous | Tier2 | Benign/Likely benign | 26.2 | 0.194 |
| **TST** | p.Ser241Leu | chr22:37011199 | rs140912272 | Heterozygous | Tier2 | Uncertain significance | 15.8 | 0.178 |
| **TTC21B** | p.Leu473Phe | chr2:165924648 | rs2163649 | Heterozygous | Tier3 | Benign | 14.1 | 0.111 |
| **UBE2A** | c.-157-36745C>T | chrX:119536898 | rs1967298 | Heterozygous | Tier2 | Not reviewed | 1.5 | — |
| **UBR5** | c.886+9G>A | chr8:102345387 | rs760900237 | Heterozygous | Tier2 | Not reviewed | 21.0 | — |
| **UCK2** | c.99+808G>T | chr1:165828740 | rs562989479 | Heterozygous | **Tier1** | Not reviewed | 6.9 | — |
| **UGT2B4** | c.721+2T>C | chr4:69495139 | rs377159596 | Heterozygous | Tier2 | Not reviewed | 23.8 | — |
| **ULBP2** | p.Ile237Leu | chr6:149947397 | rs146872141 | Heterozygous | Tier3 | Benign | 0.2 | 0.026 |
| **UQCC2** | c.138+87G>T | chr6:33711462 | rs7759182 | Heterozygous | Tier2 | Benign | 15.7 | — |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **UQCRQ** | c.*236C>T | chr5:132867818 | rs116229809 | Heterozygous | Tier2 | Uncertain significance | 4.9 | — |
| **USP47** | c.1519-181G>C | chr11:11929863 | rs140531317 | Heterozygous | Tier2 | Not reviewed | 13.4 | — |
| **VAMP1** | c.130-6C>G | chr12:6466006 | rs781012325 | Heterozygous | Tier2 | Uncertain significance | 4.9 | — |
| **VAPB** | c.*5250T>C | chr20:58449485 | rs6015274 | Heterozygous | Tier2 | Benign | 9.5 | — |
| **VPS13B** | p.Arg1740fs | chr8:99639998 | chr8:99639998 | Homozygous | Tier2 | Not reviewed | — | — |
| **VPS13B** | p.Thr1869Met | chr8:99642196 | rs117148013 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 5.6 | 0.041 |
| **VPS13B** | p.Asn2139Ser | chr8:99699894 | rs142248228 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 0.0 | 0.061 |
| **VPS13C** | p.Ser2904Leu | chr15:61911844 | rs115869241 | Heterozygous | Tier2 | Benign/Likely benign | 25.4 | 0.233 |
| **VPS13C** | c.6975+93A>G | chr15:61922304 | rs77733354 | Heterozygous | Tier2 | Benign | 0.6 | — |
| **VPS13D** | p.Ser2684= | chr1:12327709 | rs28687897 | Heterozygous | Tier2 | Benign | 6.0 | — |
| **VPS35L** | p.Thr518= | chr16:19629820 | rs144908771 | Heterozygous | **Tier1** | Likely benign | 22.5 | — |
| **VWA3B** | p.Glu372Ter | chr2:98162976 | rs62154921 | Heterozygous | **Tier1** | Benign | 46.0 | — |
| **VWF** | c.658-1399A>G | chr12:6076950 | rs7135976 | Heterozygous | Tier2 | Not reviewed | 4.7 | — |
| **WASHC4** | p.Tyr38His | chr12:105111175 | rs35294333 | Heterozygous | Tier3 | Benign | 22.8 | 0.145 |
| **WDHD1** | p.Val210Ala | chr14:55002157 | rs374497720 | Heterozygous | **Tier1** | Not reviewed | 25.8 | 0.271 |
| **WDR62** | p.Val85Met | chr19:36058855 | rs146274964 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 24.8 | 0.154 |

| Gene | Variant | Coordinate | rsID | Zygosity | Tier | ClinVar | CADD (Phred) | REVEL |
|---|---|---|---|---|---|---|---|---|
| **XDH** | p.Ile646Val | chr2:31370399 | rs17323225 | Heterozygous | Tier3 | Benign/Likely benign | 0.0 | 0.008 |
| **ZAN** | p.His1922Tyr | chr7:100773850 | rs314299 | Heterozygous | Tier2 | Benign | 18.2 | — |
| **ZAP70** | c.1737-3C>T | chr2:97739372 | rs56249179 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 14.9 | — |
| **ZBTB11** | c.1801-886G>T | chr3:101660914 | rs116667442 | Heterozygous | Tier2 | Not reviewed | 1.5 | — |
| **ZBTB20** | c.-255+46175_-255+46176del | chr3:114454210 | chr3:114454210 | Homozygous | Tier2 | Not reviewed | — | — |
| **ZFHX3** | p.Gln2759His | chr16:72794405 | rs2229288 | Heterozygous | Tier2 | Benign | 19.3 | 0.393 |
| **ZFHX3** | c.-1705_-1696dup | chr16:73891745 | chr16:73891745 | Heterozygous | Tier2 | Not reviewed | — | — |
| **ZNF808** | p.Arg494His | chr19:52554397 | rs149957912 | Heterozygous | Tier2 | Conflicting classifications of pathogenicity | 0.1 | 0.007 |
| **ZNRF3** | p.His637Arg | chr22:29050091 | rs62641746 | Heterozygous | Tier2 | Not reviewed | 0.6 | 0.024 |
| **ZRSR2** | p.Ser447_Arg448dup | chrX:15823108 | chrX:15823108 | Heterozygous | Tier3 | Likely benign | — | — |
| **ZSWIM6** | p.Ala147_Gly153del | chr5:61332694 | rs1186750536 | Heterozygous | Tier2 | Uncertain significance | — | — |
| **ZYX** | c.1024-1941C>G | chr7:143386278 | rs531543959 | Heterozygous | Tier2 | Not reviewed | 3.8 | — |